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Forecasting Electricity Consumption and CO₂ Intensity in DK1 Using Machine Learning and Time-Series Methods

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ABSTRACT

This thesis develops and evaluates machine learning models for one-hour-ahead forecasting of electricity consumption and CO₂ emission intensity in the DK1 bidding zone of Western Denmark, using data from January 2022 to December 2024. DK1 presents a structurally challenging forecasting environment: wind generation is large relative to local demand, cross-border flows across five interconnectors actively shape the system state, and the relationship between weather and both target variables is nonlinear. Four models are compared - a naïve lag-1 persistence baseline, a naïve lag-24 persistence baseline, linear regression, and a gradient boosting regressor - evaluated on a held-out test set covering the full year 2024. Feature importance and SHAP analysis are used to examine which predictors drive each model, complemented by a sensitivity analysis that removes the lag-1 target variable to reveal the contribution of exogenous predictors.

The results differ substantially between the two targets. For electricity consumption, gradient boosting achieves a test MAE of 65.13 MWh - a 36.5% improvement over the lag-1 persistence baseline and a 23.7% improvement over linear regression - corresponding to a relative error of 2.37% of average consumption. The improvement suggests that the model captures additional nonlinear temporal and system-level patterns beyond simple persistence, although the dominance of lag-1 features means that individual drivers should be interpreted cautiously. For CO₂ emission intensity, no trained model improves over the lag-1 baseline in MAE terms. The persistence model achieves a MAE of 15.52 gCO₂/kWh, while the tuned gradient boosting model reaches 15.70 gCO₂/kWh - 1.2% worse. This finding reflects the fundamental structure of CO₂ intensity in a high-renewable system: it is governed by instantaneous wind output and changes on timescales that lagged system variables cannot reliably anticipate at a one-hour horizon.

Feature importance from the full predictive models is dominated by the lag-1 target in both cases, reflecting autocorrelation structure rather than substantive system drivers. The sensitivity analysis reveals that for consumption, lagged production and exchange variables carry the most predictive weight, while for CO₂ intensity, lagged conventional generation and renewable generation emerge as dominant - consistent with DK1's wind-driven generation mix.

Keywords: *electricity consumption forecasting, CO₂ emission intensity, gradient boosting, DK1, SHAP, one-hour-ahead forecasting, high-renewable power systems*

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Denmark's electricity system has shifted substantially toward renewable energy in recent years, with renewable sources accounting for 82.1% of domestic electricity supply in 2023 (Danish Energy Agency, 2024). This has been particularly noticeable in DK1, the Western Denmark bidding zone, where a significant portion of Denmark's offshore and onshore wind capacity is located and where the system's structural characteristics make short-term forecasting both crucial and methodologically difficult.

Three structural features make DK1 a particularly relevant case for short-term forecasting in a high-renewable electricity system. First, wind generation in DK1 is large relative to local demand. The zone regularly becomes a net exporter during high wind periods and imports from Germany, Norway, and Sweden during low wind periods. These countries have significantly different generation mixes and, consequently, different CO₂ intensities. As a result, DK1's carbon intensity is more challenging to predict than in more self-contained systems since it depends on the direction and volume of cross-border flows rather than only domestic generation. Second, the relationship between weather and system outcomes is both strong and nonlinear in DK1: temperature drives heating demand in the winter and cooling demand in the summer, wind speed affects offshore and onshore generation capacity, and these two effects interact. Third, DK1's system state at any given hour reflects both local and broader regional conditions because it is involved in the Nordic and Central Western European electricity markets. Together, these features create a forecasting environment where the interactions between weather, renewable output, and cross-border flows are central and where models capable of capturing nonlinear patterns may provide advantages over purely linear benchmarks, although this must be tested empirically.

Accurate short-term forecasting of both electricity consumption (load) and CO₂ emission intensity at hourly resolution has become increasingly important for multiple stakeholders in this context. Grid operators such as Energinet must manage five active interconnectors in DK1, where wind generation can shift from surplus to deficit within hours, making reliable short-term consumption forecasts essential for reserve scheduling and cross-border flow management (Hong & Fan, 2016). Policymakers and environmental planners require CO₂ intensity forecasts to evaluate decarbonization progress, design carbon pricing mechanisms, and assess the real-time environmental impact of electricity consumption. Furthermore, as demand-side flexibility and sector coupling (such as power-to-heat systems) become more common (Lund et al., 2017), accurate forecasting makes it easier to integrate variable renewables and enables more effective demand response programs.

Despite the operational importance of these forecasts, relatively few studies have addressed both electricity consumption and CO₂ intensity forecasting simultaneously at hourly resolution using modern machine learning techniques in the Danish context. While extensive research exists on load forecasting using classical time-series methods such as ARIMA and exponential smoothing (Misiurek et al., 2025), the question of whether that advantage holds at hourly resolution in a system as renewable-intensive as DK1 is yet to be explored.

For the DK1 bidding zone, this thesis creates and assesses a data-driven forecasting framework that uses gradient boosting machine learning models to estimate hourly electricity consumption and CO₂ emission intensity. Instead of focusing on making causal connections, the emphasis is on predictive performance and determining which observable variables: weather, production mix, or cross-border exchanges drive prediction accuracy in DK1's particular operational environment.

1.2 Research Objectives and Questions

The overall aim of this thesis is to develop and evaluate machine learning-based models for short-term forecasting of hourly electricity consumption and CO₂ emission intensity in DK1 by using weather, production mix, and cross-border flows as input variables. These models will then be compared to simple linear and time-series baselines.

This aim is pursued through three specific research questions:

RQ1: How accurately can hourly electricity consumption and CO₂ emission intensity in DK1 be forecast using data-driven machine learning models based on weather, production mix, and cross-border exchange variables?

This question addresses the fundamental feasibility and performance of machine learning forecasting in the DK1 context. It establishes baseline accuracy levels and identifies the practical utility of the modeling approach for operational decision-making.

RQ2: To what extent do gradient boosting models improve forecast accuracy relative to linear regression and naïve persistence baselines for hourly electricity consumption and CO₂ emission intensity in DK1?

This question evaluates the added value of increased model complexity. By comparing gradient boosting (scikit-learn's GradientBoostingRegressor) against simple baselines—a naïve persistence model and linear regression—we assess whether the computational and conceptual overhead of machine learning methods yields meaningful performance gains in this specific forecasting context.

RQ3: Which input variables contribute most to the predictive performance of machine learning models for forecasting electricity consumption and CO₂ emission intensity in DK1?

These questions provide insight into the variables that influence forecast accuracy as well as the fundamental structure of the DK1 electrical system. We determine whether weather variables, generation mix, cross-border flows, or temporal patterns influence predictive performance and whether these drivers differ between consumption and CO₂ intensity forecasting by evaluating feature importance using both gain-based measures and SHAP (SHapley Additive exPlanations) values. However, when the lag-1 target variable is included, its dominance in feature importance obscures the contribution of all other predictors. RQ3 is therefore addressed more fully through the diagnostic no-lag-1 setup, which removes the autoregressive shortcut and forces the model to rely on exogenous variables.

1.3 Scope and Delimitations

This study focuses specifically on the DK1 bidding zone and adopts several methodological boundaries to maintain feasibility:

Geographical scope: The analysis is restricted to DK1 (Western Denmark) rather than combining DK1 and DK2 or analyzing the entire Danish system. This decision guarantees geographic consistency between system variables and weather measurements (from DMI Station 06072 in Western Denmark). Combining DK1 and DK2 could obscure relationships and lower model accuracy because of their significant differences in generating mix, consumption patterns, and interconnector designs.

Temporal scope: The study period covers January 2022 through December 2024. All three data sources - Energinet production and consumption settlement data, Energinet CO₂ emission intensity data, and DMI weather observations from Station 06072 - are publicly available and internally consistent across this period. Using recent data ensures the models reflect the current state of a system where renewables now exceed 80% of generation.

Forecasting horizon: The models produce one-hour-ahead point forecasts. Each model produces one prediction for hour $t+1$ based on the data provided at hour t . This horizon establishes which input variables are acceptable without causing data leakage and is operationally significant for real-time reserve scheduling and balancing choices. In particular, contemporaneous values of system variables, including exchange volumes, production mix, and the target variables themselves, are not known at forecast time and are therefore not included as direct inputs. Instead, the target variables capture immediate, daily, and weekly temporal dependencies by entering the models as lagged values at 1-hour, 24-hour, and 168-hour intervals. Only one-hour lags are entered for production and exchange variables. In order to approximate the use of a short-term numerical weather forecast, weather variables (temperature, wind speed, and solar radiation) are considered as known at forecast time.

Modeling approach: The thesis uses supervised machine learning methods, specifically gradient boosting through the scikit-learn's GradientBoostingRegressor algorithm, which is compared against linear regression and naïve persistence baselines. Other machine learning approaches such as deep learning (e.g., LSTM networks) (Wang et al., 2019), ensemble methods combining multiple model families, or hybrid time-series models (e.g., ARIMAX) are not fully explored in depth, although they are acknowledged as potential direction for future research.

Data coverage and limitations: Weather, production mix, and power exchanges are just a few of the major observable factors of consumption and CO₂ intensity that are together captured by the datasets. However, some relevant factors such as short-term changes in industrial activity, regulatory actions, demand-response events, or transmission grid maintenance schedules are not monitored at hourly resolution or are not made publicly available. Remaining forecast mistakes could be caused by these unseen factors. When analyzing model performance, the thesis clearly recognizes these limits and interprets forecasts as dependent on the factors that are included.

1.4 Contribution and Relevance

This thesis makes four contributions:

Empirical contribution: Using current (2022–2024) data, the thesis offers an empirical evaluation of machine learning forecasting performance for CO₂ intensity and electricity consumption in the Danish DK1 bidding zone. DK1's combination of high renewable penetration, active cross-border interconnection, and volatile CO₂ intensity makes it a more demanding forecasting environment than most systems studied in the existing literature.

Methodological contribution: The two-setup design - a full predictive model for accuracy evaluation and a diagnostic no-lag-1 model for feature importance, provides a practical approach to separating autocorrelation structure from exogenous driver effects in short-horizon electricity forecasting, where lag-1 persistence is a strong baseline.

Practical contribution: If the models achieve sufficient forecast accuracy, the findings may be of interest to grid operators such as Energinet for reserve scheduling, and to energy market participants for short-term planning.

Variable importance insights: By identifying which variables like weather conditions, renewable generation levels, cross-border flows, or temporal patterns contribute most to predictive accuracy, the thesis provides insights into the operational characteristics of the DK1 system and highlights which data inputs are most critical for forecasting applications.

The remainder of this thesis is organized as follows:

Chapter 2 (Literature Review) reviews existing research on electricity load forecasting and CO₂ emission intensity prediction, with attention to machine learning approaches, the role of weather and system variables, and methodological considerations for time-series forecasting in energy systems.

Chapter 3 (Data Preparation and Exploratory Analysis) describes the three data sources: Energinet production and consumption settlement data, Energinet CO₂ emission intensity data, and DMI meteorological observations along with the preprocessing pipeline, feature engineering steps, and exploratory analysis of temporal patterns, autocorrelation structure, and predictor-target relationships.

Chapter 4 (Modeling and Methodology) defines the forecasting problem formally, describes the train-validation-test split strategy, presents the three model families: naïve persistence, linear regression, and gradient boosting and covers hyperparameter tuning, evaluation metrics, and the feature importance and SHAP methodology.

Chapter 5 (Results) presents the empirical findings for both forecast targets, including aggregate model performance on the test set, percentage improvements relative to the lag-1 persistence baseline, feature importance and SHAP analysis for both the full predictive setup and the diagnostic no-lag-1 setup, and conditional performance analysis across hours of the day, seasons, and renewable generation levels.

Chapter 6 (Discussion) interprets the results in the context of the three research questions, compares findings to existing literature, discusses the practical implications for grid operators and the limitations imposed by the modeling assumptions, and reflects on the role of model complexity in this forecasting setting.

Chapter 7 (Conclusion) summarizes the key findings, highlights the main contributions, and suggests directions for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing research relevant to forecasting electricity consumption and CO₂ emission intensity in high-renewable power systems. The review is structured in five main sections: first, an overview of electricity load forecasting methods, from classical statistical approaches to modern machine learning techniques; second, a review of CO₂ emission intensity forecasting in power systems; third, a discussion of key methodological considerations including feature selection, validation strategies, and model interpretability; fourth, identification of the research gap that this thesis addresses; and finally, a summary that connects the literature to the present study.

2.2 Electricity Load Forecasting: Methods and Evolution

2.2.1 Classical Time-Series Approaches

Traditional load forecasting has relied heavily on statistical time-series models. ARIMA (AutoRegressive Integrated Moving Average) and its seasonal variant SARIMAX (Misiurek et al., 2025) have been widely applied due to their ability to capture temporal dependencies and seasonal patterns in electricity demand. Exponential smoothing methods, including Holt-Winters seasonal models, have also seen extensive use for their simplicity and computational efficiency.

These classical approaches excel at capturing autocorrelation - the strong relationship between current load and recent historical values - which is a dominant feature of electricity consumption patterns. Daily, weekly, and yearly seasonality are typically well-represented by these models. However, classical time-series methods face limitations when incorporating exogenous variables such as weather conditions, economic indicators, or generation mix. While SARIMAX extends ARIMA to include exogenous regressors, it assumes linear relationships between these variables and the target, which may not adequately capture the complex, nonlinear interactions present in modern power systems with high renewable penetration. These limitations are relevant for DK1 specifically, where the nonlinear relationship between temperature and consumption and the interaction between renewable generation and cross-border flows are unlikely to be captured by a linear time-series model.

2.2.2 Machine Learning Approaches to Load Forecasting

Recent research has increasingly turned to machine learning methods for electricity load forecasting, motivated by their ability to model nonlinear relationships and automatically capture interaction effects between variables.

Support Vector Regression (SVR) has demonstrated strong performance in regional load forecasting. (Chen et al., 2023) applied linear regression, support vector regression, and random forest regression to monthly electricity demand forecasting in Guangdong Province. Their results showed that SVR and random forest achieved the best predictive performance among the tested models.

However, Chen et al.'s analysis was conducted at monthly resolution, which is considerably coarser than the hourly forecasting pursued in the present thesis. Monthly aggregation smooths out the short-term variability that is critical for operational grid management, and the relationships between predictors and consumption may differ substantially at hourly versus monthly timescales.

Tree-based ensemble methods, particularly Random Forest and gradient boosting algorithms, have gained prominence in recent load forecasting applications. These methods construct multiple decision trees and aggregate their predictions, allowing them to capture complex nonlinear patterns and variable interactions without requiring explicit feature engineering of interaction terms. The superior performance of tree-based methods in capturing these complex relationships provides a strong motivation for employing gradient boosting in the present study. However, most of this evidence comes

from systems with lower renewable penetration and coarser temporal resolution than DK1, leaving the performance of gradient boosting at hourly resolution in a high-renewable European context as an open empirical question.

2.2.3 The Role of Weather Variables

Across the literature, weather variables - particularly temperature, wind speed, and solar radiation emerge as critical predictors of electricity consumption. Temperature drives heating demand in winter and cooling demand in summer, creating strong seasonal patterns. In high-renewable systems like DK1, wind speed and solar radiation are also important because they directly influence the availability of renewable generation, which in turn affects the need for conventional generation, cross-border imports, and storage dispatch. (Chen et al., 2023) found weather variability to be a key contributor to seasonal fluctuations in electricity demand, alongside socio-economic development. This finding aligns with broader literature emphasizing the weather-dependency of both demand and supply in modern power systems.

2.3 CO₂ Emission Intensity Forecasting in Power Systems

2.3.1 CO₂ Intensity Dynamics in High-Renewable Systems

In power systems with significant renewable generation, CO₂ emission intensity (typically measured in grams of CO₂ per kilowatt-hour consumed) exhibits substantial short-term variability. Periods of high wind and solar generation are often associated with lower CO₂ intensity, although realized intensity also depends on demand, imports, exports, and market dispatch (Milligan et al., 2011). Conversely, during periods of low renewable output, conventional thermal plants must meet demand, increasing carbon intensity. Additionally, cross-border electricity exchanges introduce further complexity: importing electricity from a neighboring region with high coal generation increases domestic carbon intensity, while exporting surplus renewable power reduces it (Fell et al., 2021).

Denmark exemplifies these dynamics. With over 80% renewable electricity in recent years, Danish CO₂ intensity can range from near-zero during high-wind periods to several hundred grams per kWh when thermal generation is needed or high-carbon electricity is imported. These dynamics make DK1 a particularly challenging forecasting environment and motivate the inclusion of conventional generation, renewables, lagged wind and exchange variables as predictors.

2.3.2 Machine Learning for CO₂ Intensity Forecasting

Several studies have applied machine learning to CO₂ intensity forecasting, though the literature is less mature than for load forecasting.

CarbonCast, developed by (Maji et al., 2023), is a notable example of a machine learning framework for regional carbon intensity forecasting. CarbonCast uses historical generation mix data, weather

variables, and time-of-day features to produce multi-day forecasts of grid carbon intensity across multiple regions. The authors demonstrated that a CNN-LSTM hierarchical neural network could achieve accurate multi-day carbon intensity forecasts, with performance significantly exceeding simple baselines. (Maji et al., 2023) emphasized the value of feature engineering-creating lagged variables, temporal indicators, and aggregated renewable generation features-to improve forecast accuracy.

(Leerbeck et al., 2020) developed a short-term forecasting framework for average and marginal CO₂ intensity in Denmark's DK2 bidding zone. Their approach combined linear regression with extensive feature selection and ARIMA-based residual correction. Starting with 473 candidate variables-including production, demand, imports, weather, and variables from neighboring zones- (Leerbeck et al., 2020) employed LASSO regression and forward feature selection to reduce the feature set to approximately 30 variables. This feature selection step was crucial for avoiding overfitting and improving generalization performance.

(Leerbeck et al., 2020) then built three linear models with different specifications (including interaction terms in some variants) and combined their predictions using a softmax weighted average. Finally, they applied ARIMA to the residuals, creating a hybrid ARIMAX model that captured both the exogenous variable relationships (via linear regression) and the remaining temporal autocorrelation (via ARIMA). Their results showed normalized root mean squared error (NRMSE) ranging from 0.095 to 0.183 for average CO₂ intensity forecasting in DK2, demonstrating reasonable accuracy.

(Leerbeck et al., 2020) found that marginal emissions in DK2 were significantly influenced by conditions in neighboring zones, validating the importance of including cross-border flow variables in models for interconnected systems like DK1.

The study demonstrated that systematic feature selection is critical when starting with a large candidate feature set, that rolling time-aware validation is necessary to avoid overoptimistic performance estimates, and that conditions in neighboring zones significantly influence CO₂ intensity in interconnected systems. The existing CO₂ intensity forecasting literature leaves a clear gap: no published study has applied gradient boosting with systematic baseline comparison to hourly CO₂ forecasting in DK1.

2.4 Methodological Considerations

Beyond specific forecasting methods, several methodological considerations are critical for sound time-series forecasting research in energy systems.

2.4.1 Feature Engineering and Selection

Feature selection - deciding which of many candidate features to include in the final model-helps prevent overfitting, improve computational efficiency, and enhance model interpretability. (Leerbeck et al., 2020) demonstrated the value of systematic feature selection by reducing 473 variables to 30. In machine learning contexts, formal feature selection methods include LASSO regularization, recursive feature elimination, or permutation-based importance filtering. However, gradient boosting models have built-in mechanisms for handling large feature sets through regularization and can provide feature importance rankings that serve a similar purpose to explicit feature selection.

2.4.2 Time-Aware Validation Strategies

For hourly electricity data with strong autocorrelation, random train-test splits allow the model to observe future values during training which is a form of leakage that produces overoptimistic estimates and does not reflect operational forecasting conditions.

(Leerbeck et al., 2020) employed rolling cross-validation with eight sequential rounds, where each round used past data for training and future data for testing. This approach ensures that the evaluation mimics real-world forecasting scenarios where only historical data is available.

2.4.3 Model Interpretability and Explainability

As machine learning models become more complex, understanding which features drive predictions becomes increasingly important, both for scientific insight and for building stakeholder trust. Feature importance analysis helps identify the key drivers of forecast accuracy and can reveal whether a model is learning sensible relationships or exploiting spurious correlations.

Traditional tree-based models provide gain-based feature importance scores, which measure how much each feature contributes to reducing prediction error across all trees. However, gain-based importance has limitations: it is biased toward high-cardinality features, does not show the direction of effects, and cannot reveal interaction effects clearly.

SHAP (SHapley Additive exPlanations) values, introduced by (Lundberg & Lee, 2017), provide a more sophisticated approach to model interpretability. SHAP values are based on cooperative game theory and assign each feature a contribution to the prediction for each individual observation. Compared to gain-based importance, SHAP provides directional, observation-level attribution that is not biased toward high-cardinality features, making it better suited for interrogating what the gradient boosting models in this study have actually learned.

2.5 Research Gap and Positioning of This Thesis

While extensive literature exists on both electricity load forecasting and CO₂ emission intensity prediction, several gaps remain that this thesis addresses:

Hourly resolution in the Danish context: (Leerbeck et al., 2020) addressed CO₂ intensity forecasting in DK2 but relied on linear models and did not extend to DK1. Very few studies have applied gradient boosting to hourly CO₂ intensity forecasting in DK1, which differs from DK2 in generation mix and interconnector configuration.

Combined load and CO₂ forecasting: Load forecasting and carbon intensity forecasting are typically studied in isolation. The extent to which the same feature set explains both targets simultaneously and whether the dominant drivers differ between them has not been addressed thoroughly in the Danish context.

Gradient boosting with interpretability in high-renewable European systems: Existing ML applications to CO₂ intensity forecasting (Maji et al., 2023) focus on multi-day horizons in US systems. Whether gradient boosting with SHAP-based interpretability performs well at hourly resolution in a European high-renewable context remains unexamined.

Systematic baseline comparison: Many ML forecasting studies compare only against one baseline or against other ML methods. A three-way comparison - naïve persistence, linear regression, and gradient boosting - that quantifies the marginal value of model complexity at each step is relatively uncommon in the Danish CO₂ or load forecasting literature.

Recent data reflecting current system state: Studies using pre-2020 data may not reflect the dynamics of a system where renewables exceed 80% of generation. The 2022–2024 period used here captures a system state not covered by prior work.

2.6 Summary

The key gap this thesis addresses is the absence of a systematic, baseline-anchored evaluation of gradient boosting for both consumption and CO₂ intensity forecasting at hourly resolution in DK1, a system whose renewable penetration and interconnector dependence make it a distinctively demanding forecasting environment.

The literature does not uniformly support the superiority of gradient boosting over simpler approaches. Performance gains from ML methods vary substantially across temporal resolutions, system types, and feature sets. Several studies find that well-specified linear models with appropriate feature selection perform comparably to more complex methods, particularly when the nonlinear structure of the data is moderate. For CO₂ intensity forecasting specifically, the evidence base is thin only a handful of studies exist, and most operate at different resolutions or in different system contexts than DK1. Whether the performance advantages documented in the load forecasting literature transfer to CO₂ intensity forecasting at hourly resolution in high-renewable European systems is an empirical question that this thesis addresses directly.

The present thesis addresses this gap directly by developing a gradient boosting framework for hourly forecasting of electricity consumption and CO₂ intensity in DK1, employing time-aware validation, systematic baseline comparison, and SHAP-based interpretability analysis to answer the three research questions posed in Chapter 1.

CHAPTER 3: DATA PREPARATION AND EXPLORATORY ANALYSIS

3.1 Introduction

This chapter describes the data sources, preprocessing procedures, feature engineering approaches, and exploratory analysis conducted to prepare the dataset for modeling. The analysis focuses on the DK1 bidding zone in Western Denmark and covers the period from January 1, 2022, to December 31, 2024. Three primary data sources are integrated: electricity production and consumption data from Energinet, CO₂ emission intensity data from Energinet, and meteorological observations from the Danish Meteorological Institute (DMI). The chapter proceeds as follows: Section 3.2 describes the three data sources in detail, Section 3.3 outlines the preprocessing steps applied to each dataset, Section 3.4 explains the data integration and merging process, Section 3.5 presents the feature engineering approach, Section 3.6 provides exploratory data analysis and visualization of key patterns.

3.2 Data Sources

3.2.1 Production and Consumption Settlement Data

The primary dataset for electricity production, consumption, and cross-border exchanges is sourced from Energinet's open data platform, specifically from the Production Consumption Settlement dataset available at energidataservice.dk. This dataset provides hourly settlement data for the Danish electricity system, separated by bidding zone (DK1 for Western Denmark and DK2 for Eastern Denmark). The dataset is publicly available and maintained by Energinet, Denmark's transmission system operator.

The Production Consumption Settlement dataset contains 28 variables covering various aspects of the electricity system:

Generation sources: The dataset includes separate columns for central power generation, local power generation, commercial power, offshore wind generation (split into plants below and above 100 MW capacity), onshore wind generation (split by capacity thresholds at 50 kW), hydropower, solar power generation (categorized by capacity: below 10 kW, between 10–40 kW, and above 40 kW), and unknown production sources.

Cross-border electricity exchanges: Six interconnector flow variables capture electricity exchanges between DK1 and neighboring regions: Norway (ExchangeNO_MWh), Sweden (ExchangeSE_MWh), Germany (ExchangeGE_MWh), the Netherlands (ExchangeNL_MWh), Great Britain (ExchangeGB_MWh), and the Great Belt connection to DK2 (ExchangeGreatBelt_MWh). Positive values indicate imports into DK1, while negative values indicate exports from DK1.

Consumption and grid variables: The dataset includes gross consumption in megawatt-hours (GrossConsumptionMWh), grid losses in transmission networks, grid losses in interconnectors, grid losses in distribution networks, and power-to-heat consumption.

Temporal identifiers: Each observation includes both UTC timestamps (HourUTC) and Danish local time timestamps (HourDK), as well as a PriceArea identifier indicating the bidding zone (DK1 or DK2).

For this analysis, only observations corresponding to the DK1 bidding zone were retained. The raw dataset as downloaded contained 26,304 hourly observations spanning the full three-year period from January 1, 2022, through December 31, 2024, with no missing hours. All 28 variables were retained for subsequent feature engineering and analysis.

3.2.2 CO₂ Emission Intensity Data

CO₂ emission intensity data is also sourced from Energinet's data platform. This dataset provides estimates of the average CO₂ intensity of electricity consumed in each bidding zone, measured in grams of CO₂ per kilowatt-hour (g CO₂/kWh). The CO₂ intensity figure represents the average carbon content of electricity consumed in the zone at each point in time, weighted across the domestic generation mix and adjusted for the carbon content of cross-border imports and exports.

The raw CO₂ emission dataset is provided at 5-minute resolution, resulting in significantly higher temporal granularity than the hourly production and consumption data. The raw dataset contained 315,611 observations covering the period from January 1, 2022, to December 31, 2024, for DK1. Each observation includes:

- Minutes5UTC: Timestamp in UTC at 5-minute intervals
- Minutes5DK: Timestamp in Danish local time
- PriceArea: Bidding zone identifier (DK1 or DK2)
- CO₂ Emission: Estimated CO₂ emission intensity in grams per kWh

As with the production and consumption data, only observations corresponding to DK1 were retained. The CO₂ emission values in the raw dataset were formatted using European decimal conventions (comma as decimal separator), requiring conversion to standard floating-point format before numerical analysis. After filtering for DK1, the dataset was aggregated from 5-minute to hourly resolution by computing the mean CO₂ intensity across the twelve 5-minute intervals within each hour. This aggregation approach ensures consistency with the hourly resolution of the production and consumption data while preserving the central tendency of CO₂ emissions within each hour. The aggregated dataset contained 26,304 hourly observations, matching the temporal coverage of the production and consumption data.

3.2.3 Meteorological Data

Weather data is obtained from the Danish Meteorological Institute (DMI), which provides open access to meteorological observations from weather stations across Denmark. The data is available through DMI's open data portal at <https://www.dmi.dk/vejrkarkiv>, with documentation available at <https://www.dmi.dk/friedata/dokumentation/>.

DMI Station 06072 was selected as a Western Denmark weather station and is used as a regional weather proxy for DK1. Because DK1 covers a broad area, this station should not be interpreted as fully representing all weather variation across DK1. The choice of a Western Denmark weather station supports spatial consistency between weather conditions and the electricity system variables, which are specific to the DK1 region. Western and Eastern Denmark can experience different weather conditions simultaneously due to their geographical separation, making it important to use weather observations from the same region as the electricity system data.

The DMI weather dataset provides observations at 10-minute resolution and includes three key meteorological variables:

- Temperature (°C): Air temperature measured in degrees Celsius, which drives heating and cooling demand
- Wind speed (m/s): Wind speed in meters per second, which directly determines wind power generation capacity and influences grid balancing requirements.
- Solar radiation (W/m²): Solar irradiance in watts per square meter, which affects solar power generation potential and influences cooling demand during daylight hours

The raw weather dataset contained 153,437 observations at 10-minute intervals covering the period from January 1, 2022, to December 31, 2024. Each observation includes timestamps in both UTC and Danish local time. As with the CO₂ data, the weather observations were aggregated to hourly resolution to match the temporal granularity of the electricity system data. For each hour, the mean value of temperature, wind speed, and solar radiation across the six 10-minute observations within that hour was computed. The aggregated weather dataset contained 26,304 hourly observations, maintaining complete temporal alignment with the production, consumption, and CO₂ datasets.

The three meteorological variables were selected because they represent the primary weather drivers of electricity consumption and renewable generation in the Danish context. Temperature influences both heating demand (during winter months) and cooling demand (during summer months), wind speed is directly related to wind power output, and solar radiation drives solar photovoltaic generation and affects daytime demand patterns.

3.3 Data Preprocessing

3.3.1 CO₂ Emission Intensity Preprocessing

The CO₂ emission intensity data required several preprocessing steps to prepare it for integration with the other datasets. The raw data was loaded from a semicolon-separated CSV file (CO₂Emis.csv) containing 315,611 rows at 5-minute resolution.

Timestamp handling: The raw dataset contained two timestamp columns: Minutes5UTC (timestamps in Coordinated Universal Time) and Minutes5DK (timestamps in Danish local time). For consistency with the other datasets and to avoid complications associated with daylight saving time transitions, only the UTC timestamp column was retained (Minutes5UTC). This column was converted from string format to pandas datetime objects with UTC timezone awareness.

Decimal format conversion: The CO₂ emission values (CO₂Emission column) were stored as text strings using European decimal notation, where commas serve as decimal separators (e.g., "63,000000" representing 63.0 g CO₂/kWh). These values were converted to standard floating-point numbers by replacing comma separators with periods and then applying pandas' `to_numeric()` function with error handling to ensure robust conversion.

Geographic filtering: The analysis focuses exclusively on the DK1 bidding zone. Therefore, only observations where PriceArea equals DK1 were retained.

Missing timestamp handling: A timestamp continuity check was performed by creating a complete 5-minute UTC timestamp range from the first to the last observation and comparing it with the observed timestamps. This check identified 37 missing 5-minute timestamps. These missing timestamps were inserted through reindexing to create a regular 5-minute time series. Missing CO₂Emission values introduced during this step were filled using time-based interpolation, with forward and backward filling used as safeguards for any remaining missing values.

Temporal aggregation: To align with the hourly resolution of the production and consumption data, the 5-minute CO₂ observations were aggregated to hourly resolution. The timestamp column was set as the index, and *pandas'* *resample()* function was applied with a 1-hour frequency and mean aggregation. This approach computes the average CO₂ intensity across all 5-minute observations within each hour, providing a representative hourly CO₂ value. The resulting dataset was renamed to use a consistent `datetime_utc` column and a simplified `co2_intensity` target variable name.

After preprocessing, the CO₂ dataset contained 26,304 hourly observations with three columns: `datetime_utc` (hourly timestamp in UTC), `co2_intensity` (CO₂ emission intensity in g/kWh), and `PriceArea` (always "DK1").

3.3.2 Production and Consumption Settlement Preprocessing

The production and consumption settlement data required minimal preprocessing because it was already provided in hourly format and filtered for the DK1 bidding zone. The data was loaded from an CSV file (`ProductionConsumptionSettlement.csv`) containing 26,304 rows and 28 columns.

Timestamp standardization: The dataset contained two timestamp columns: `HourUTC`, representing Coordinated Universal Time (UTC), and `HourDK`, representing Danish local time (CET/CEST). Consistent with the preprocessing approach applied to the CO₂ emission intensity and weather datasets, only the UTC timestamp column (`HourUTC`) was retained as the common temporal reference and renamed to `datetime_utc`. The timestamp column was converted to *pandas* `datetime` format with UTC timezone awareness to ensure temporal consistency across all datasets and to avoid complications associated with daylight saving time transitions. The `HourDK` column was not used for merging operations.

Geographic filtering: The dataset included a `PriceArea` column identifying whether each observation belonged to DK1 or DK2. A verification step confirmed that all observations corresponded exclusively to DK1 (Western Denmark). After this validation, the `PriceArea` column was removed from the preprocessed dataset because it contained only a single unique value and therefore did not provide additional modeling information.

Missing value handling: A systematic missing-value assessment was conducted across all variables. The hourly timestamp sequence was complete, with no missing hourly observations across the study period from January 1, 2022, through December 31, 2024. However, the variable `ExchangeGB_MWh`, representing electricity exchange between Denmark and Great Britain, contained missing values in earlier observations. These missing values were interpreted as zero flows because the Viking Link interconnector became operational only toward the end of the study period. Therefore, missing values in `ExchangeGB_MWh` were replaced with zeros, representing the absence of physical electricity exchange during those hours.

Numeric data verification: Several numeric variables in the raw dataset were stored using European decimal notation, where commas were used as decimal separators. These values were converted to floating-point format by replacing commas with periods and applying numeric conversion procedures. This preprocessing step ensured that all electricity generation, consumption, grid-loss, and cross-border exchange variables could be used consistently for statistical analysis, visualization, and feature engineering.

Dataset integrity verification: After preprocessing, the production and consumption settlement dataset retained all 26,304 hourly observations. The cleaned dataset used `datetime_utc` as the common timestamp variable and contained electricity production, consumption, grid-loss, power-to-heat, and cross-border exchange variables for subsequent analysis and integration with the CO₂ emission intensity and weather datasets.

3.3.3 Weather Data Preprocessing

The meteorological data from DMI Station 06072 required several preprocessing steps before integration with the electricity system and CO₂ datasets.

Timestamp handling: The raw weather dataset contained 153,437 observations at 10-minute resolution. Two timestamp columns were provided: `datetime_utc` (UTC time) and `datetime_dk` (Danish local time). Consistent with the other datasets, the UTC timestamp column was retained as the common temporal reference. The `datetime_utc` column was converted from ISO 8601 string format to pandas datetime format with UTC timezone awareness. The `datetime_dk` column was removed to avoid complications related to daylight saving time transitions during merging and resampling operations.

Missing timestamp handling: A timestamp continuity check was performed by comparing the observed timestamps with a complete expected 10-minute UTC timestamp range spanning the study period. This process identified 4,388 missing timestamps in the raw weather dataset. Missing timestamps were inserted through reindexing to create a regular 10-minute time series.

The reindexing process introduced missing values in the meteorological variables. These missing values were filled using time-based interpolation, followed by forward and backward filling as safeguards for any remaining missing observations. After preprocessing, all weather variables contained complete observations across the study period.

Temporal aggregation: The cleaned 10-minute weather observations were aggregated to hourly resolution using pandas' `resample("1h").mean()` function. For each hour, the arithmetic mean of temperature, wind speed, and solar radiation was calculated from the six 10-minute observations within the hour. Mean aggregation was selected because it provides a representative summary of sustained weather conditions relevant for electricity demand and renewable generation forecasting.

Variable verification: The three meteorological variables (temperature, `wind_speed`, and `solar_radiation`) were verified to be stored as floating-point numeric variables with appropriate physical units.

Outlier assessment: The aggregated hourly weather variables were examined for statistical outliers using the interquartile range (IQR) method. Temperature produced 38 outlier observations, wind speed produced 404, and solar radiation produced 2,882. These observations were retained because they corresponded to plausible meteorological variability rather than clear measurement errors. Temperature exhibited seasonal variability, wind speed showed a moderately right-skewed distribution typical of atmospheric processes, and solar radiation displayed near-zero nighttime values with higher daytime summer peaks.

The final preprocessed weather dataset contained 26,304 hourly observations and four columns: `datetime_utc`, temperature, `wind_speed`, and `solar_radiation`.

3.4 Data Integration and Merging

After preprocessing each of the three datasets individually, they were integrated into a single unified dataset through a sequence of inner joins on the `datetime_utc` timestamp column.

3.4.1 Merging Strategy

The merging process proceeded in two steps to ensure data integrity and facilitate validation at each stage:

Step 1: Merging production/consumption with CO₂ data. The production and consumption settlement dataset (26,304 rows, 36 columns) were merged with the CO₂ emission intensity dataset (26,304 rows, 3 columns) using an inner join on `datetime_utc`. An inner join was selected to ensure that only observations with matching timestamps in both datasets were retained, excluding any potential rows where timestamp alignment might be imperfect. Because both datasets covered the identical time period (January 1, 2022, through December 31, 2024) at hourly resolution with no missing hours, the inner join preserved all 26,304 observations. The merged dataset contained 37 columns: `datetime_utc`, the 35 production/consumption variables and `co2_intensity` (present in both source datasets).

Step 2: Adding weather data. The partially merged dataset from Step 1 was then merged with the preprocessed weather dataset (26,304 rows, 4 columns) using an inner join on `datetime_utc`. Again, because all three datasets covered the same complete hourly time series from 2022 through 2024 with no gaps, the inner join preserved all 26,304 observations. The final merged dataset contained 40 columns: `datetime_utc`, the 35 production/consumption variables, `co2_intensity`, the 3 weather variables (temperature, wind_speed, solar_radiation).

3.4.2 Merged Dataset Structure

The merged dataset has the following structure:

- Temporal dimension: 26,304 hourly observations spanning January 1, 2022, 00:00 UTC through December 31, 2024, 23:00 UTC, representing three complete calendar years
- Geographic scope: All observations correspond to the DK1 (Western Denmark) bidding zone
- Variables: 40 columns comprising temporal identifiers, production and generation variables, consumption and grid loss variables, cross-border exchange flows, the CO₂ emission intensity target, and meteorological observations

The successful completion of inner joins with no data loss confirms perfect temporal alignment across all three data sources. Each hourly timestamp from 2022–2024 is represented in all three source datasets, ensuring that the modeling framework will have complete information on production, consumption, CO₂ intensity, and weather conditions for every hour in the study period.

3.5 Feature Engineering

Feature engineering was conducted in multiple stages. Aggregated electricity generation and cross-border exchange variables were first constructed within the production and consumption settlement dataset prior to dataset integration. After merging the production and consumption, CO₂ emission intensity, and weather datasets, additional lagged variables and time-based calendar features were created to support the forecasting framework.

The purpose of feature engineering was to reduce dimensionality, improve interpretability, and create operationally meaningful predictors representing broader electricity system conditions in DK1.

3.5.1 Aggregated Generation Features

To reduce dimensionality and create more interpretable predictors, related electricity generation sources were aggregated into higher-level categories.

Total wind generation (`TotalWind_MWh`) was calculated as the sum of all wind generation categories:

- `OffshoreWindLt100MW_MWh`
- `OffshoreWindGe100MW_MWh`

- OnshoreWindLt50kW_MWh
- OnshoreWindGe50kW_MWh

This aggregation creates a single predictor representing total hourly wind power generation in DK1. Higher wind generation is generally associated with lower CO₂ intensity because renewable electricity contributes a larger share of the generation mix.

Total solar generation (TotalSolar_MWh) was calculated as the sum of all solar photovoltaic generation categories:

- SolarPowerLt10kW_MWh
- SolarPowerGe10Lt40kW_MWh
- SolarPowerGe40kW_MWh
- SolarPowerSelfConMWh

This feature captures total hourly solar electricity production across residential, commercial, and utility-scale installations. Solar generation exhibits strong diurnal and seasonal patterns and is expected to influence both electricity demand patterns and CO₂ intensity.

Total renewable generation (TotalRenewables_MWh) was calculated as the sum of TotalWind_MWh, TotalSolar_MWh, and HydroPowerMWh. This variable represents the total renewable electricity generation in DK1 during each hour and is expected to be strongly associated with variations in CO₂ intensity.

Total conventional and other non-renewable generation (TotalConventional_MWh) was calculated as the sum of:

- CentralPowerMWh
- LocalPowerMWh
- CommercialPowerMWh
- LocalPowerSelfConMWh
- UnknownProdMWh

This feature captures electricity generation from central power plants, local generation facilities, commercial production units, self-consumed local production, and unidentified production sources

3.5.2 Cross-Border Flow Aggregations

Cross-border electricity exchange variables were aggregated to summarize DK1's overall import and export behavior.

Net exchange (NetExchange_MWh) was calculated as the sum of all six interconnector flow variables:

$$\text{NetExchange_MWh} = \text{ExchangeNO_MWh} + \text{ExchangeSE_MWh} + \text{ExchangeGE_MWh} + \text{ExchangeNL_MWh} + \text{ExchangeGB_MWh} + \text{ExchangeGreatBelt_MWh}$$

Positive values indicate net electricity imports into DK1, while negative values indicate net exports from DK1.

Total imports (TotalImports_MWh) were calculated as the sum of all positive cross-border exchange values. This variable represents the total magnitude of electricity imported into DK1 during each hour.

Total exports (TotalExports_MWh) were calculated as the absolute sum of all negative exchange values. The resulting variable therefore represents export magnitude as a positive quantity.

These exchange variables summarize the interaction between DK1 and neighboring electricity systems and are expected to contribute to both electricity consumption and CO₂ intensity forecasting performance.

3.5.3 Total Production

Total electricity production (TotalProduction_MWh) was calculated as:

$$\text{TotalProduction_MWh} = \text{TotalRenewables_MWh} + \text{TotalConventional_MWh}$$

This variable represents the total hourly electricity generation within DK1 across renewable and conventional production sources.

3.5.4 Lagged Features

To capture temporal autocorrelation and enable the models to learn from recent historical values, lagged versions of the two target variables were created:

CO₂ intensity lags:

- `co2_lag_1`: CO₂ intensity from 1 hour ago (t-1)
- `co2_lag_24`: CO₂ intensity from 24 hours ago (t-24), capturing daily seasonality
- `co2_lag_168`: CO₂ intensity from 168 hours ago (t-168), capturing weekly seasonality (7 days × 24 hours)

Consumption lags:

- `cons_lag_1`: Gross consumption from 1 hour ago (t-1)
- `cons_lag_24`: Gross consumption from 24 hours ago (t-24)
- `cons_lag_168`: Gross consumption from 168 hours ago (t-168)

The selection of lag-1, lag-24, and lag-168 is motivated by both established conventions in time-series forecasting for electricity systems and empirical evidence from autocorrelation analysis (presented in Section 3.6.2). These lags capture immediate persistence (lag-1), daily patterns (lag-24), and weekly patterns (lag-168).

Lagged variables were created using pandas' *shift()* function. The first observation (t=0) has no lag-1 value, the first 24 observations have no lag-24 values, and the first 168 observations have no lag-168 values. These missing values resulting from the lagging process are addressed during dataset cleaning by removing all incomplete observations.

3.5.5 Time-Based Features

To capture recurring human activity patterns that drive electricity demand, a set of calendar features was extracted from the Danish local timestamp (Europe/Copenhagen timezone). The dataset timestamps are maintained in Coordinated Universal Time (UTC) to ensure a continuous and consistent temporal structure; however, time-based features were derived from Danish local time to better reflect real-world consumption behavior. Daylight Saving Time (DST) transitions introduce repeated or missing hours in local time, but this does not affect the integrity of the underlying UTC timestamps.

The following time-based features were created:

- hour: Hour of the day (0–23), derived from Danish local time
- day_of_week: Day of the week (0 = Monday through 6 = Sunday)
- month: Calendar month (1–12)
- is_weekend: Binary indicator equal to 1 for Saturday and Sunday, 0 otherwise

Season encoding: A season variable was first derived from the month variable using the following mapping: winter (December, January, February), spring (March, April, May), summer (June, July, August), and autumn (September, October, November). The season variable was then one-hot encoded into three binary columns - season_spring, season_summer, and season_winter - with autumn serving as the dropped reference category. This encoding avoids multicollinearity, which is particularly important for linear regression models. Although tree-based models such as gradient boosting are not affected by multicollinearity, the same encoding was retained across all models to ensure consistency and comparability of feature effects.

3.5.6 Lagged Production and Exchange Features

Contemporaneous production and exchange variables - such as TotalWind_MWh, TotalRenewables_MWh, TotalProduction_MWh, and NetExchange_MWh - cannot be used directly as model inputs because they are not known at prediction time (see Chapter 4 for details on the forecasting setup). To preserve the predictive value of these variables while avoiding data leakage, lag-1 versions of each were created:

- wind_lag1: Total wind generation from the previous hour (t-1)
- solar_lag1: Total solar generation from the previous hour (t-1)
- renewables_lag1: Total renewable generation from the previous hour (t-1)
- conventional_lag1: Total conventional generation from the previous hour (t-1)
- production_lag1: Total production from the previous hour (t-1)
- imports_lag1: Total imports from the previous hour (t-1)
- exports_lag1: Total exports from the previous hour (t-1)
- netexchange_lag1: Net exchange from the previous hour (t-1)

These lagged production and exchange features allow the model to incorporate information about the most recent known energy mix when generating forecasts. For instance, wind_lag1 captures the wind generation level in the hour immediately preceding the forecast target, providing a strong signal for both CO₂ intensity (through renewable penetration) and consumption (through system balancing dynamics). A lag of one hour was selected because it represents the most recent information realistically available at prediction time and avoids the look-ahead bias that would arise from using contemporaneous values.

3.5.7 Final Dataset Characteristics

After feature engineering, the full dataset contains 62 variables. During the modeling stage, a leakage-controlled model-ready dataset was constructed by removing all contemporaneous production and exchange variables that would not be available at prediction time. This process retained only variables

that are either known in advance (time-based features, weather conditions) or based on past observed values (lagged features and lag targets). After this step and after removing all rows with missing values introduced by lag construction, the model-ready dataset contains 28 columns and 26,135 complete observations, broken down as follows:

1. Temporal identifiers (2): `datetime_utc`, `datetime_dk`
2. Target variables (2): `GrossConsumptionMWh`, `co2_intensity`
3. Weather variables (3): `temperature`, `wind_speed`, `solar_radiation`
4. Time-based features (7): `hour`, `day_of_week`, `month`, `is_weekend`, `season_spring`, `season_summer`, `season_winter`
5. CO₂ intensity lag features (3): `co2_lag_1`, `co2_lag_24`, `co2_lag_168`
6. Consumption lag features (3): `cons_lag_1`, `cons_lag_24`, `cons_lag_168`
7. Lagged production and exchange features (8): `wind_lag1`, `solar_lag1`, `renewables_lag1`, `conventional_lag1`, `production_lag1`, `imports_lag1`, `exports_lag1`, `netexchange_lag1`

The first 169 observations contain missing values due to the combination of lag-168 and lag-1 production features, and are excluded after the `dropna()` step, yielding 26,135 usable observations ($26,304 - 169 = 26,135$).

The target variables for forecasting are:

- `GrossConsumptionMWh`: Hourly electricity consumption in DK1 (in megawatt-hours)
- `co2_intensity`: Hourly CO₂ emission intensity in DK1 (in grams of CO₂ per kilowatt-hour)

3.6 Exploratory Data Analysis

Exploratory data analysis was conducted to understand temporal patterns, validate feature engineering decisions, explore relationships between predictors and targets, and assess data distributions. The analysis focuses on identifying patterns that will inform model selection and feature importance interpretation.

3.6.1 Temporal Patterns and Seasonality

Time-series visualization. Figure 3.1 presents the hourly electricity consumption and CO₂ emission intensity in DK1 across the full three-year period from 2022 through 2024. Electricity consumption exhibits clear seasonal variation, with higher average consumption during winter months (October through March) compared to summer months (May through August). This seasonality reflects increased heating demand during colder periods. Within each year, consumption shows pronounced day-to-day and week-to-week variability, driven by changes in weather conditions, economic activity patterns, and weekend versus weekday differences.

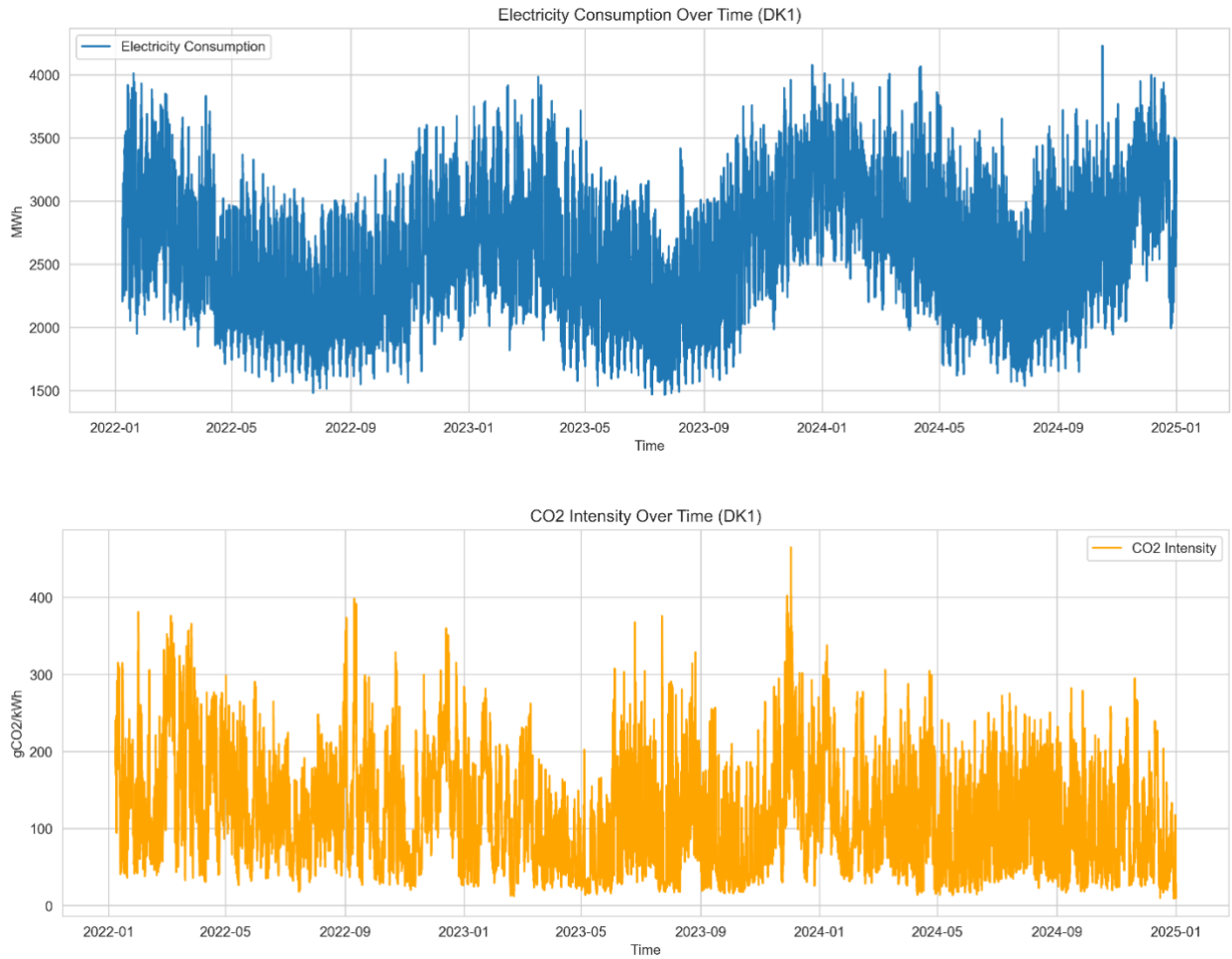


Figure 3.1. Hourly electricity consumption (top) and CO₂ emission intensity (bottom) in DK1 from January 2022 through December 2024. Consumption shows seasonal patterns with higher values in winter months, while CO₂ intensity exhibits high variability driven by renewable generation availability.

CO₂ emission intensity displays substantial short-term variability, ranging from near-zero values during periods of high renewable generation to over 400 grams per kilowatt-hour during periods when conventional thermal generation or high-carbon electricity imports are needed to meet demand. Unlike consumption, CO₂ intensity does not show consistent seasonal patterns, instead exhibiting high-frequency fluctuations driven by the variability of wind and solar generation. Extended periods of low CO₂ intensity (below 50 g/kWh) are visible throughout the dataset, corresponding to sustained high-wind conditions when renewable generation displaces conventional sources. Conversely, sharp spikes in CO₂ intensity occur during low-wind, low-solar periods when thermal generation must compensate for reduced renewable output.

Daily patterns. Figure 3.2 displays average hourly consumption and CO₂ intensity patterns across the 24 hours of the day, computed by averaging across all observations at each hour (00:00 through 23:00) in the dataset.

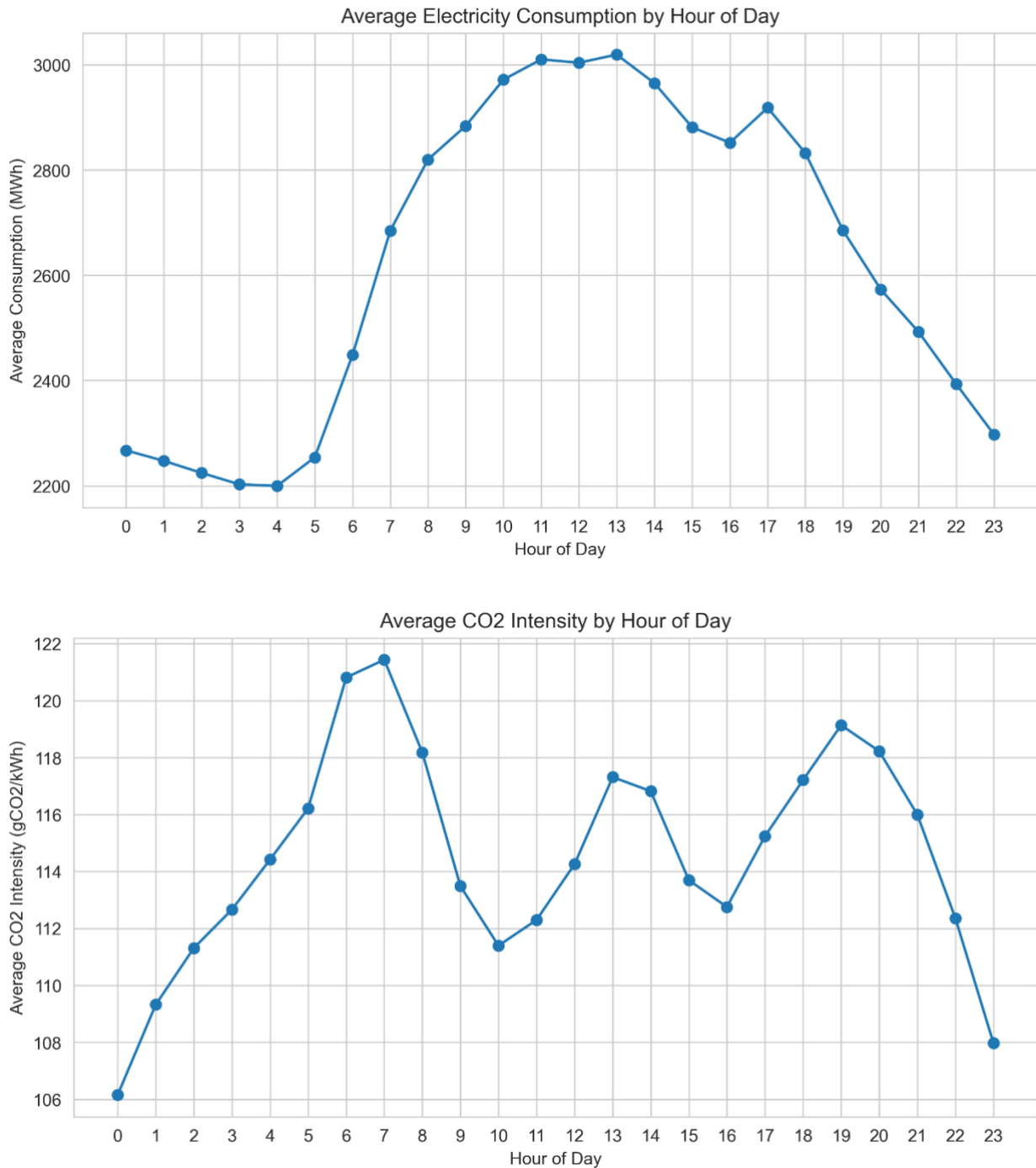
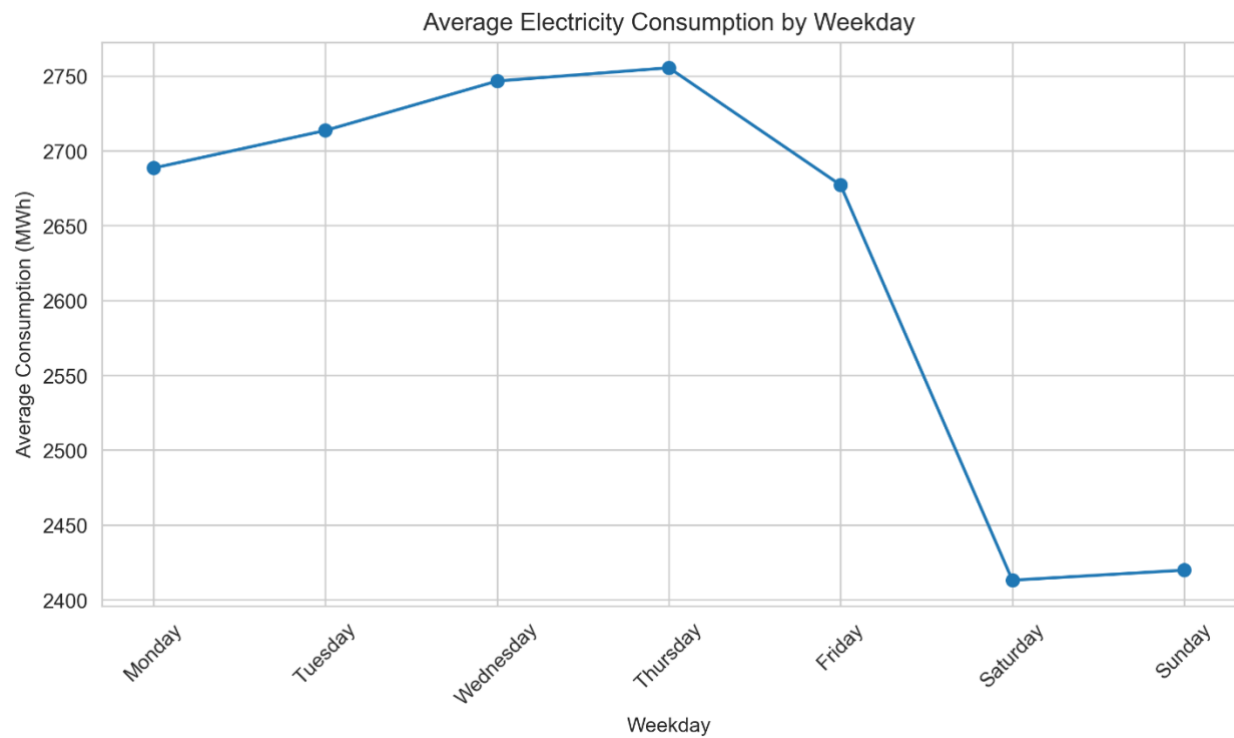


Figure 3.2. Average hourly patterns for electricity consumption (top) and CO₂ emission intensity (bottom). Consumption shows clear diurnal cycles with morning and evening peaks, while CO₂ intensity exhibits more complex patterns influenced by solar generation.

Electricity consumption shows a pronounced diurnal cycle. Consumption is lowest during the early morning hours (02:00–05:00), reflecting reduced demand when most residential, commercial, and industrial activity is minimal. Consumption begins rising around 06:00 as morning activities commence, reaching a first peak around 09:00–12:00 during daytime working hours. Consumption remains elevated through the afternoon and reaches a second peak in the evening hours around 17:00–19:00, driven by residential heating, lighting, cooking, and appliances as people return home from work. Consumption then declines through the late evening and night.

CO₂ intensity shows a more complex daily pattern. Intensity tends to be higher during morning and evening peak consumption hours when additional conventional generation may be dispatched to meet demand. However, the pattern is heavily influenced by solar generation during daylight hours: periods of high solar radiation (midday) often coincide with lower CO₂ intensity if solar generation displaces fossil fuel sources. The high variability in the CO₂ pattern reflects the strong dependence on weather-driven renewable generation, which varies significantly from day to day.

Weekly patterns. Figure 3.3 presents average consumption and CO₂ intensity by day of week, revealing clear weekday versus weekend differences.



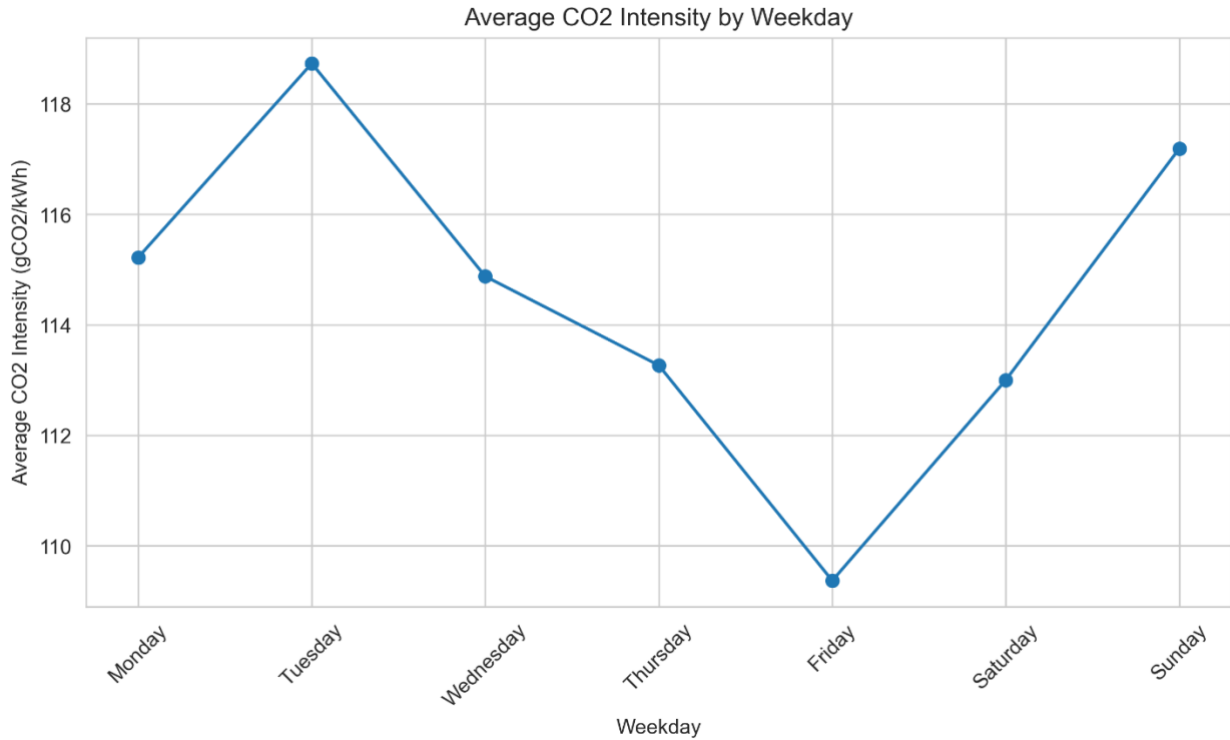


Figure 3.3. Average consumption (top) and CO₂ emission intensity (bottom) by day of week. Consumption is notably higher on weekdays compared to weekends, reflecting industrial and commercial activity patterns. CO₂ intensity shows weaker weekly variation.

Electricity consumption is notably higher on weekdays (Monday through Friday) compared to weekends (Saturday and Sunday). This pattern reflects the contribution of industrial and commercial electricity demand, which is concentrated on working days. The drop in consumption from Friday to Saturday and the recovery from Sunday to Monday are particularly evident, highlighting the influence of the weekly work cycle on aggregate electricity demand.

CO₂ intensity shows smaller and less consistent differences across days of the week. Intensity is highest at the start of the week, peaking around Tuesday, before declining steadily through Wednesday and Thursday. The lowest average CO₂ intensity occurs on Friday, suggesting relatively cleaner electricity production toward the end of the working week. This pattern then reverses over the weekend: CO₂ intensity rises again on Saturday and reaches near-peak levels on Sunday, which contradicts the simple assumption that lower weekend demand leads to lower carbon intensity. This finding highlights that CO₂ intensity is driven primarily by renewable generation availability - particularly wind - rather than demand levels and does not follow a simple weekday-weekend pattern.

3.6.2 Autocorrelation Analysis

To empirically validate the selection of lagged features (lag-1, lag-24, and lag-168), autocorrelation function (ACF) plots were generated for both target variables. The ACF measures the correlation between a time series and its own lagged values, revealing the strength and persistence of temporal dependencies.

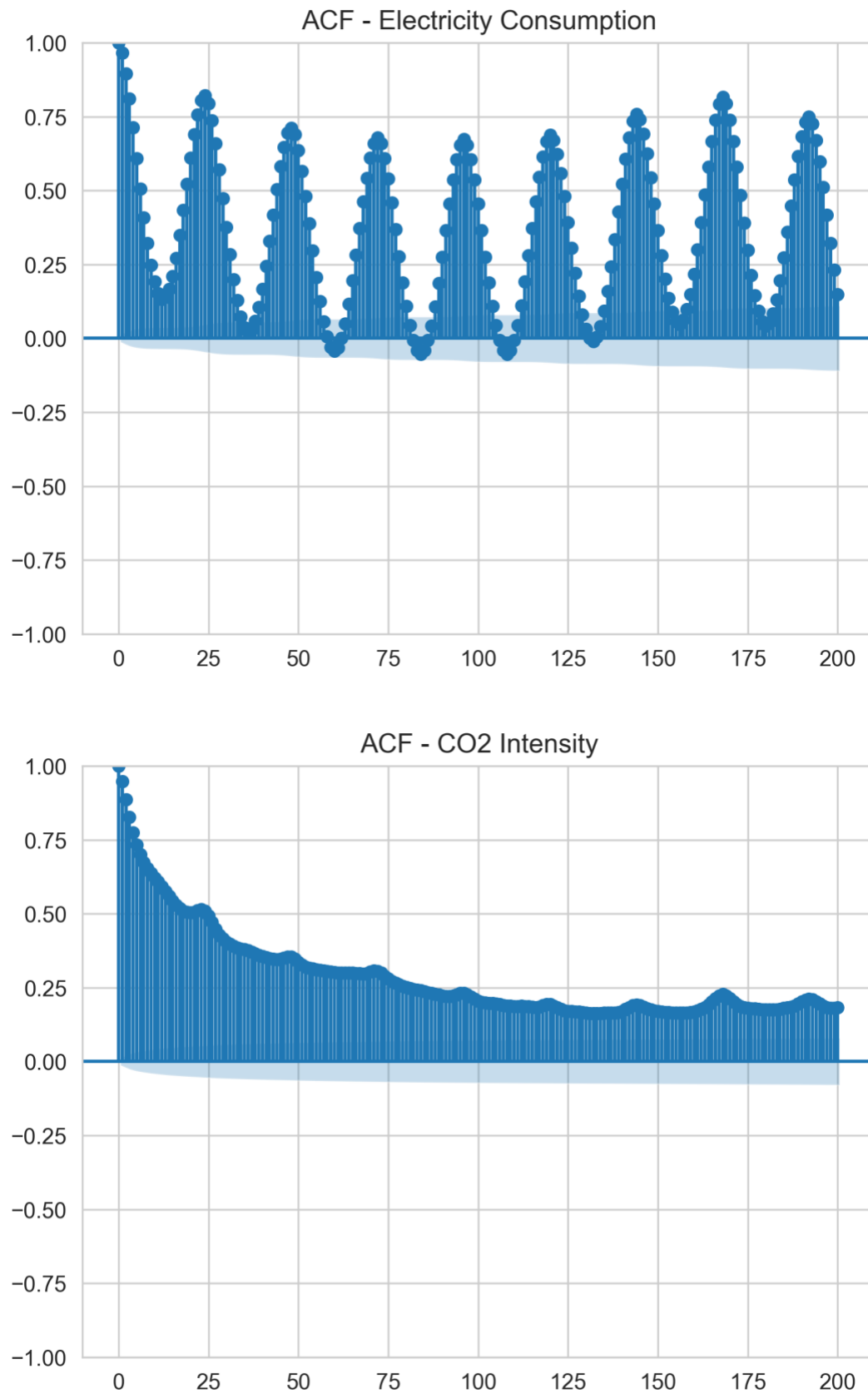


Figure 3.4. Autocorrelation function (ACF) for electricity consumption (top) and CO₂ emission intensity (bottom) across 200 hourly lags. Strong autocorrelation at lags 1, 24, and 168 for electricity consumption empirically justifies the inclusion of lagged features in the forecasting models.

Consumption autocorrelation. The ACF for gross consumption confirms very strong autocorrelation at short lags: the correlation between consumption at time t and consumption at time $t-1$ is 0.965, indicating that consumption in one hour is highly predictive of consumption in the next hour. This strong lag-1 autocorrelation justifies the inclusion of `cons_lag_1` as a predictor.

The ACF also reveals clear periodic patterns. A prominent peak occurs at lag 24, with autocorrelation of 0.822, indicating that consumption at the same hour yesterday is strongly correlated with consumption today. This 24-hour cyclical pattern reflects the daily consumption cycle identified in Section 3.6.1. The persistence of significant autocorrelation at multiples of 24 (lags 24, 48, 72, etc.) confirms the daily seasonality in consumption. Additionally, a secondary peak is visible at lag 168 (7 days $\times$ 24 hours), with autocorrelation of 0.815, corresponding to weekly seasonality. The empirical evidence of strong autocorrelation at lags 1, 24, and 168 confirms that the selected lagged features are data-driven and well-motivated by the temporal structure of consumption.

CO₂ intensity autocorrelation. The ACF for CO₂ emission intensity reveals a different temporal structure. While lag-1 autocorrelation is high at 0.947 - indicating strong short-term persistence in CO₂ intensity from hour to hour - the autocorrelation decays substantially more rapidly than for consumption. At lag 24, autocorrelation drops to 0.510 (moderate), and at lag 168 it falls to only 0.227 (weak). This pattern of smooth exponential decay, without the strong recurring seasonal peaks visible in the consumption ACF, indicates that CO₂ intensity does not exhibit a dominant daily or weekly seasonal cycle. Rather, CO₂ intensity is primarily driven by the variable availability of renewable generation - particularly wind - which does not follow predictable weekly cycles.

The contrast between the two target variables has direct implications for forecasting. For electricity consumption, lagged values at lags 1, 24, and 168 are all highly informative, and time-based features (hour of day, day of week) are likely to be strong predictors. For CO₂ intensity, the lag-1 value is similarly informative, but the weak autocorrelation at lag 168 suggests that additional explanatory variables - in particular renewable generation proxies and weather conditions - carry proportionally more weight. These findings provide strong empirical justification for including the full set of lagged features and the lagged production variables described in Section 3.5.6.

The numerical ACF values at key lags are summarized in Table 3.1 below.

Table 3.1. Autocorrelation function values at key lags for the two target variables.

Variable	Lag 1	Lag 24	Lag 168
GrossConsumptionMWh	0.965	0.822	0.815
co2_intensity	0.947	0.510	0.227

3.6.3 Predictor-Target Relationships

To provide an initial overview of linear relationships between key variables, a Pearson correlation matrix was computed for the two target variables and a selected set of predictors. Figure 3.5 presents the resulting heatmap.

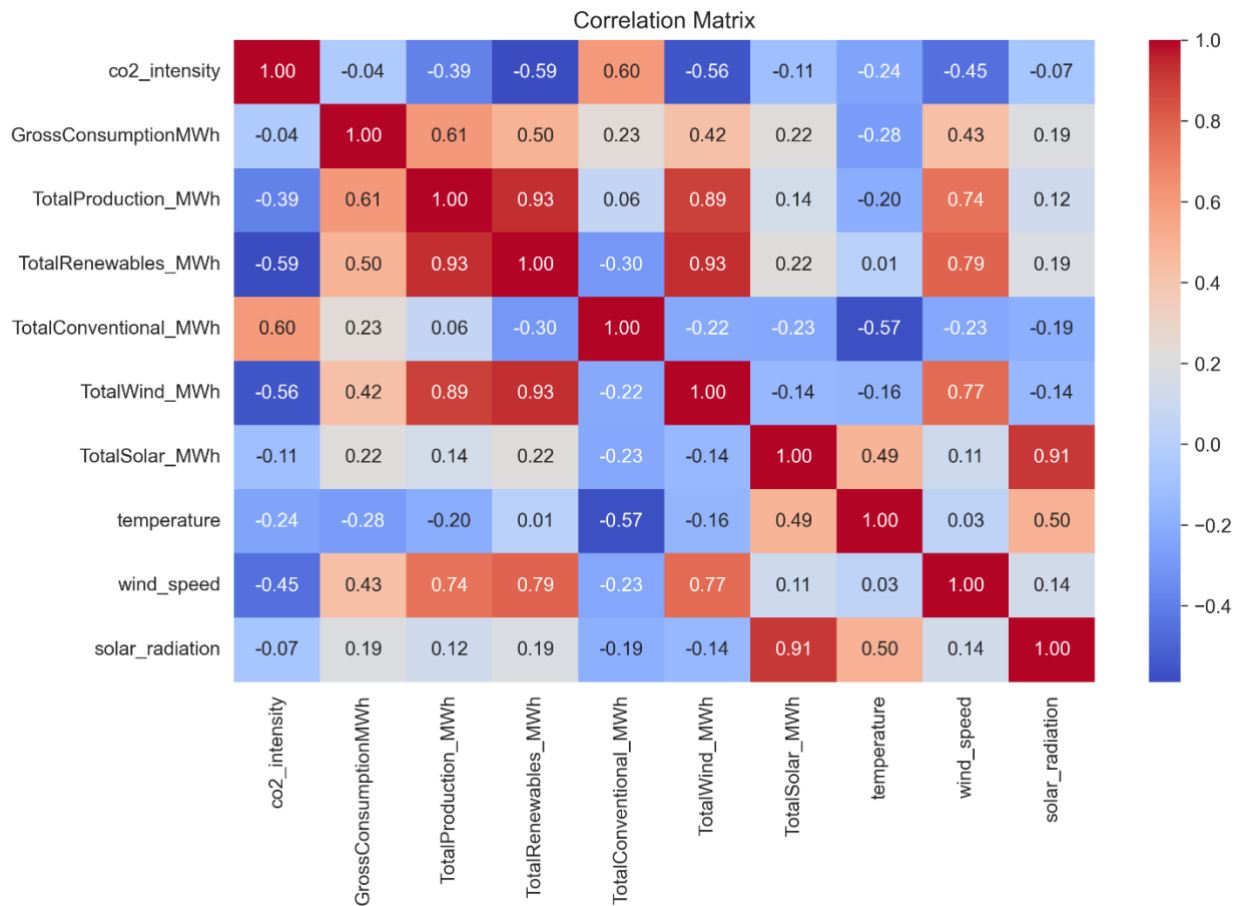


Figure 3.5. Pearson correlation matrix for key variables in the model-ready dataset. Values range from -1 (perfect negative correlation) to +1 (perfect positive correlation). Strong negative correlations with CO₂ intensity are visible for total wind and total renewables, while gross consumption shows moderate positive correlation with temperature extremes.

The matrix confirms several expected relationships. CO₂ intensity is strongly negatively correlated with total wind generation and total renewables, reflecting the well-established role of renewable energy in reducing carbon intensity. Gross consumption shows a moderate positive correlation with total production and a weaker relationship with weather variables. The correlation matrix also reveals that total wind, total renewables, and total production are highly intercorrelated, which is expected given their definitional overlap. These findings provide initial justification for the predictor selection and motivate the more detailed nonlinear analysis in the scatter plots that follow.

To assess the nature of relationships between key predictors and the target variables, bivariate scatter plots were created for selected variable pairs.

Temperature and consumption. Figure 3.6 displays the relationship between temperature and gross consumption. The scatter plot reveals a U-shaped (nonlinear) relationship. Consumption is elevated at both low temperatures (below approximately 5°C) and high temperatures (above approximately 20°C), while consumption is lowest at moderate temperatures (around 10–15°C).

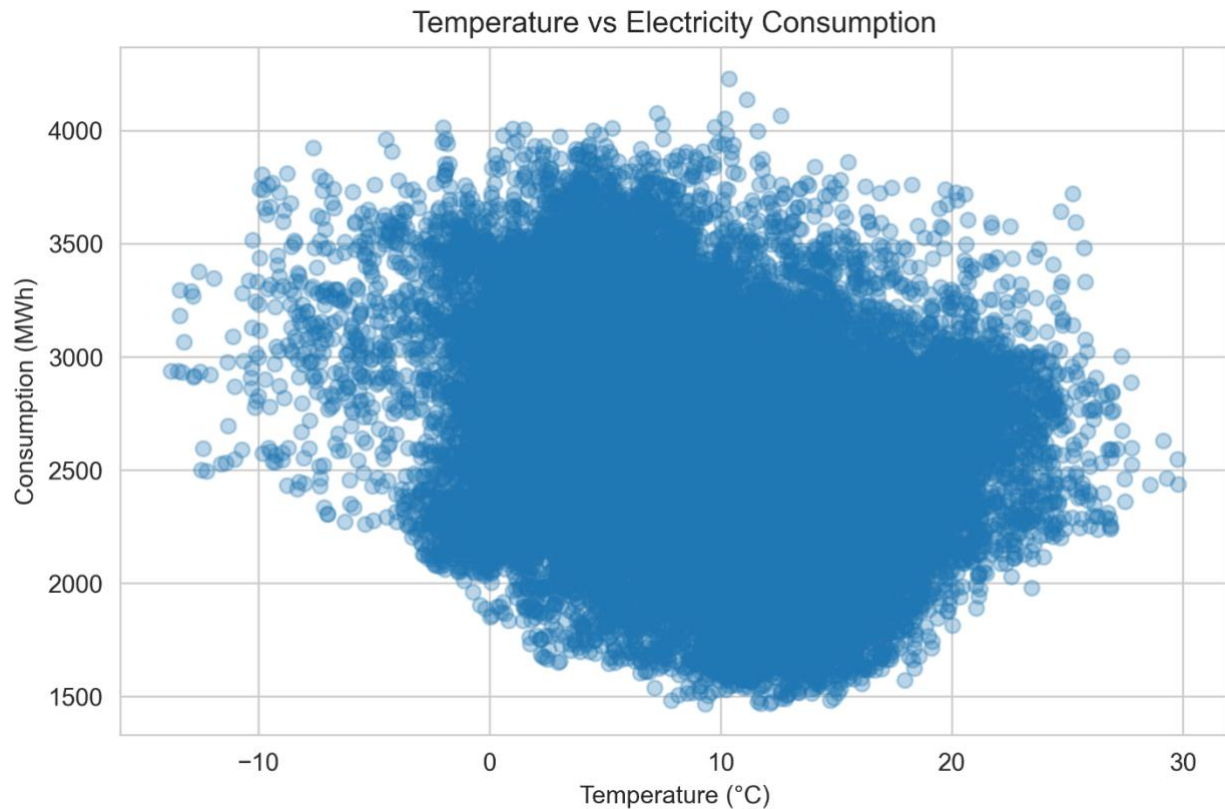


Figure 3.6. Relationship between temperature and electricity consumption. The pattern indicates increased consumption at both low temperatures (heating demand) and high temperatures (cooling demand), with minimum consumption at moderate temperatures around 12–15°C. This nonlinear relationship motivates the use of gradient boosting models.

This U-shaped pattern is consistent with the dual role of temperature in driving electricity demand. At low temperatures, electric heating systems increase consumption, particularly in Denmark where district heating is supplemented by electric heating in many buildings. At high temperatures, cooling and air conditioning systems drive increased consumption during warm periods. The presence of a strong nonlinear relationship between temperature and consumption provides clear motivation for the use of machine learning models such as gradient boosting, which can automatically capture such nonlinearities without requiring explicit specification of polynomial or interaction terms.

Wind generation and CO₂ intensity. Figure 3.7 shows the relationship between total wind generation and CO₂ emission intensity. The scatter plot reveals a strong inverse (negative) relationship: as wind generation increases, CO₂ intensity decreases. When wind generation is low (below 500 MWh), CO₂ intensity is highly variable and often exceeds 200 g/kWh, reflecting the need for conventional thermal generation or imports from high-carbon sources to meet demand. As wind generation increases above 1000 MWh, CO₂ intensity declines sharply. When wind generation exceeds 2000 MWh, CO₂ intensity frequently falls to very low levels, suggesting periods when renewable generation fully or nearly fully meets DK1's electricity demand.

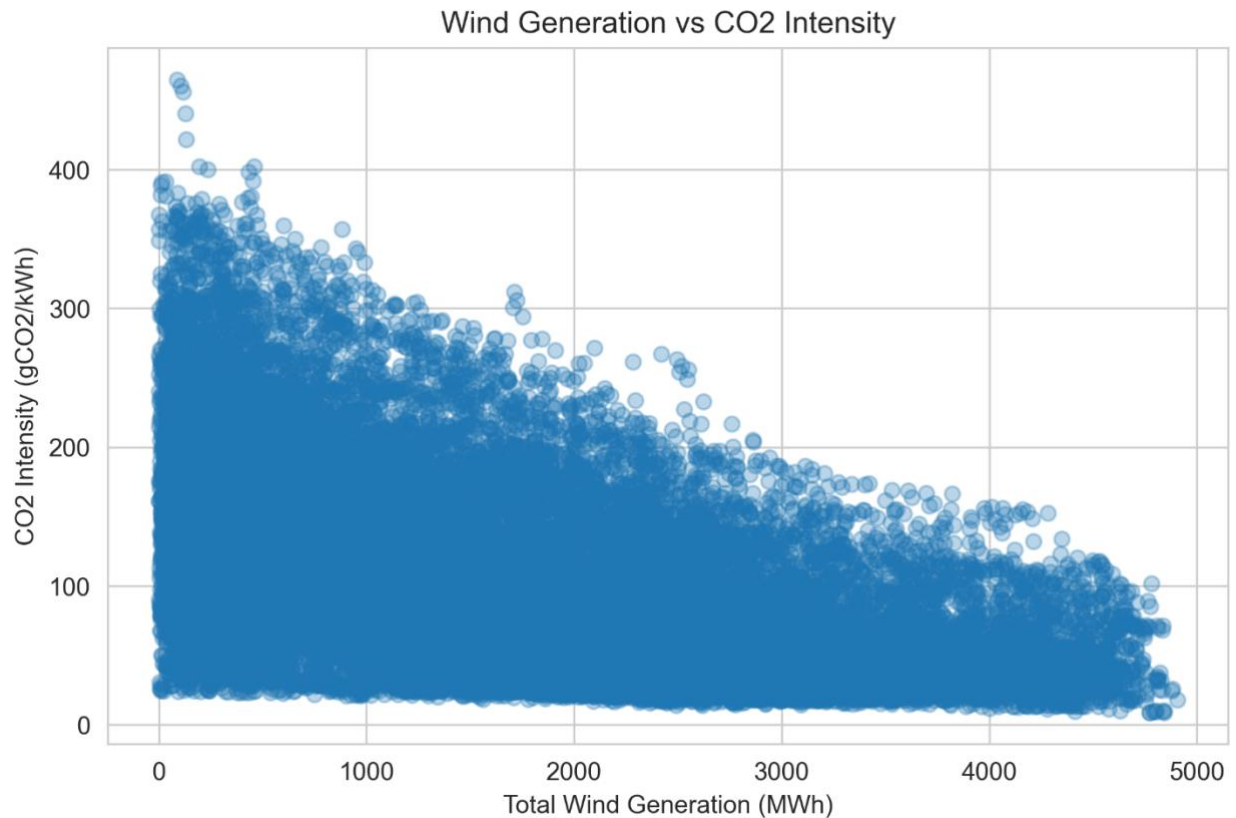


Figure 3.7. Relationship between total wind generation and CO₂ emission intensity. The strong inverse relationship demonstrates that higher wind generation is associated with lower carbon intensity.

The relationship is not perfectly linear: the steepest decline in CO₂ intensity occurs as wind generation increases from low to moderate levels, with diminishing marginal reductions in CO₂ intensity at very high wind generation levels (above 2500 MWh). This nonlinearity likely reflects the interaction between wind generation, consumption, and cross-border flows: even at very high wind generation, some conventional generation or imports may still be dispatched for grid stability or economic reasons. The strong negative relationship between wind generation and CO₂ intensity confirms the critical role of renewable energy in reducing carbon emissions and supports the inclusion of conventional generation, lagged wind and renewables features as key predictors in the CO₂ intensity forecasting models.

3.6.4 Data Distributions

Consumption distribution. Figure 3.8 (top panels) presents the distribution of gross consumption through histograms. The distribution is approximately normal with a slight right skew. The mean consumption is approximately 2,630 MWh per hour, with most observations falling between 2,200 and 3000 MWh. The right skew reflects occasional periods of very high consumption, often corresponding to cold winter days with peak heating demand. There are very few extreme low values, indicating that consumption rarely drops below 1,500 MWh even during low-demand periods.

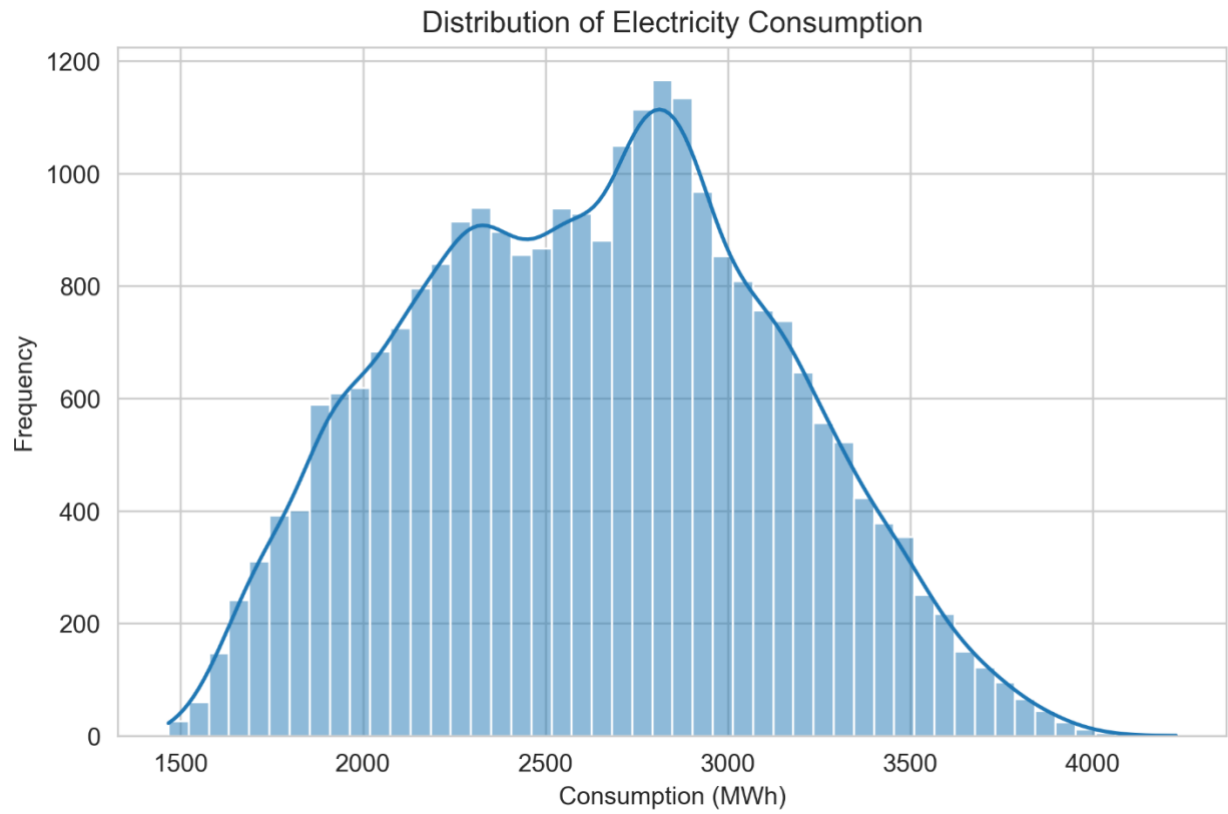


Figure 3.8. Distribution of electricity consumption shown as histograms with density curves. Consumption is approximately normally distributed with slight right skew.

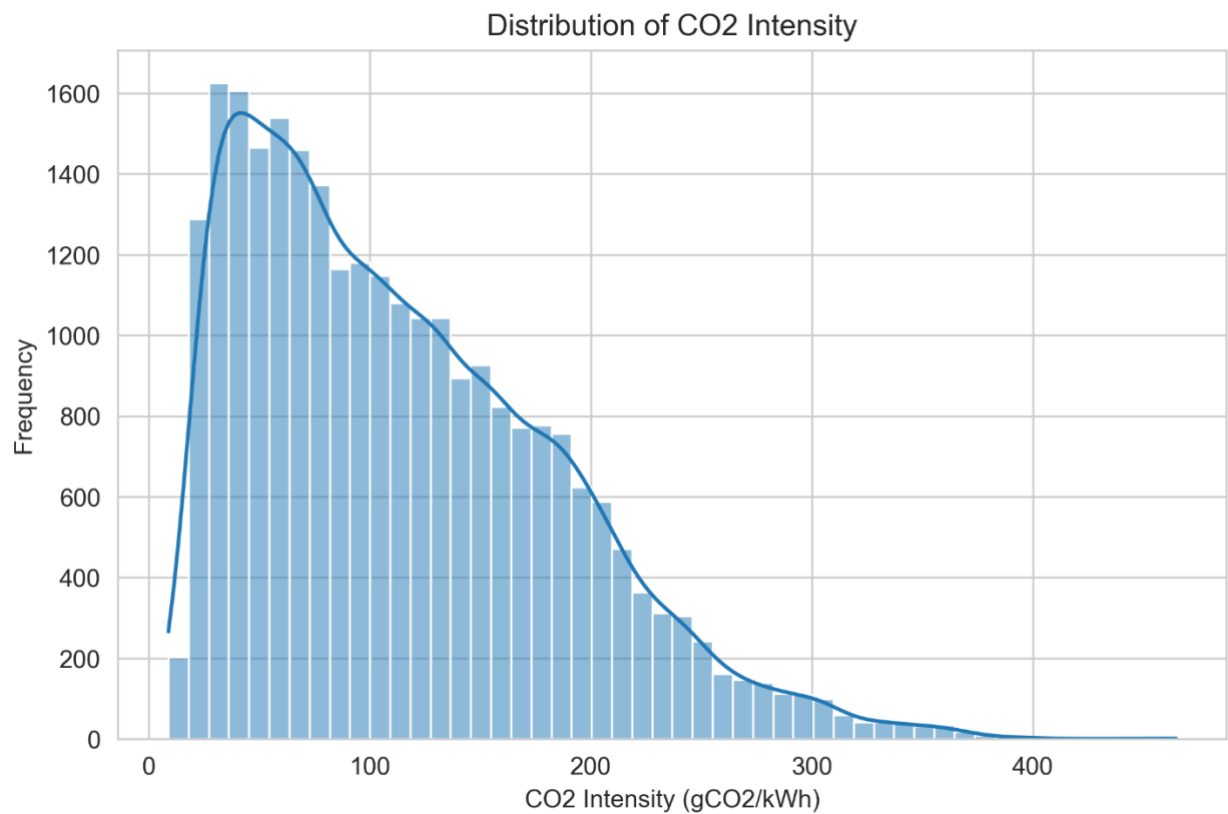


Figure 3.9. Distribution of CO₂ emission intensity shown as histograms with density curves. CO₂ intensity exhibits a right-skewed distribution with a large cluster of low values and a long upper tail.

The approximately normal distribution of consumption is advantageous for many modeling approaches, including linear regression, and suggests that standard regression loss functions (mean squared error, mean absolute error) proved appropriate for consumption forecasting.

CO₂ intensity distribution. Figure 3.9 shows the distribution of CO₂ emission intensity. Unlike consumption, CO₂ intensity exhibits a wide distribution with substantial right skew. A large proportion of observations are concentrated at very low CO₂ intensity values (0–50 g/kWh), while a long tail extends toward higher values, with some observations exceeding 400 g/kWh. The mean CO₂ intensity is approximately 115 g/kWh, but the distribution is highly asymmetric.

The right-skewed distribution reflects two distinct operational regimes in the DK1 electricity system: periods of high renewable generation (low CO₂ intensity) and periods when conventional generation or high-carbon imports are necessary (moderate to high CO₂ intensity). The asymmetry and variability of the CO₂ distribution suggest that CO₂ forecasting proved more challenging than consumption forecasting, as the target variable exhibits greater variability and a less regular distribution. These characteristics further support the use of flexible machine learning models capable of handling non-Gaussian target distributions.

3.6.5 Boxplot Analysis

To summarize the distributional properties of the two target variables, boxplots were constructed for gross consumption and CO₂ emission intensity, providing a compact representation of central tendency, spread, and outlier structure across all 26,136 hourly observations. The boxplots are constructed from the dataset prior to the creation of lag-1 production features, which yields 26,136 observations. The subsequent creation of lagged production variables and removal of incomplete rows via `dropna()` reduces this to the 26,135 observations used in modeling.

Consumption distribution. Figure 3.10 presents the boxplot for gross electricity consumption. The median sits close to 2,600 MWh, and the interquartile range spans approximately 2,300 to 3,000 MWh, indicating that the middle 50 percent of hourly consumption values are concentrated within a moderate range. The whiskers extend to 1.5 times the interquartile range at both ends, capturing the full spread of typical consumption values. Only a small number of outliers appear above the upper whisker, corresponding to episodes of unusually high demand during cold winter periods.

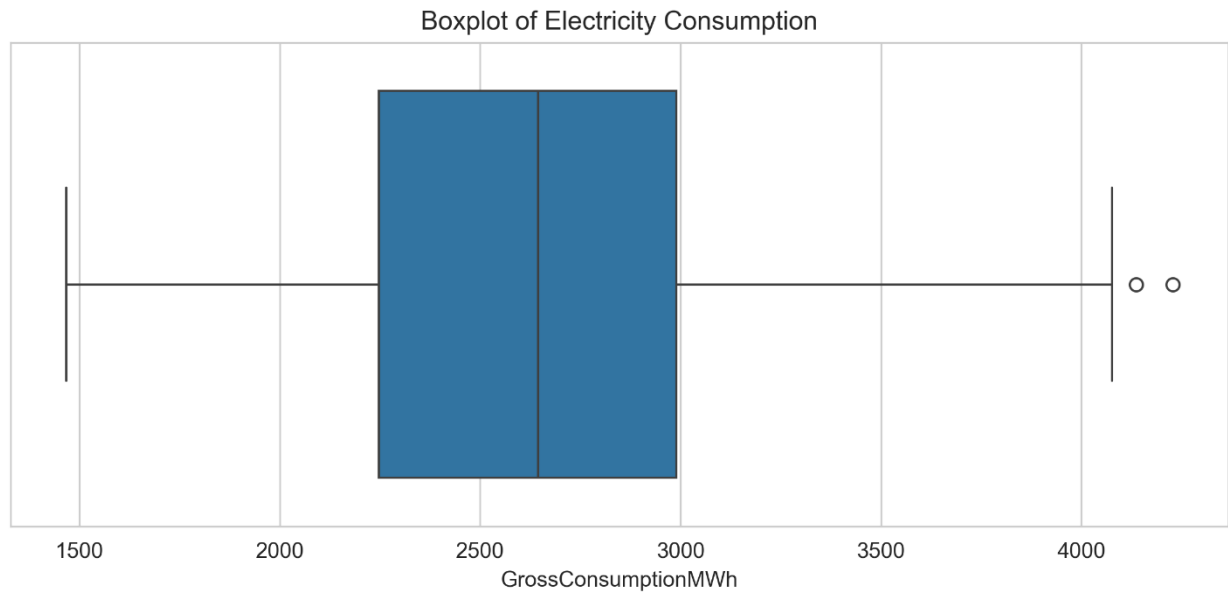


Figure 3.10. Boxplot of electricity consumption. The box represents the interquartile range (IQR), the horizontal line indicates the median, and whiskers extend to 1.5 times the IQR. The distribution is compact and approximately symmetric, with only a small number of high-value outliers corresponding to peak winter demand periods.

CO₂ intensity distribution. Figure 3.11 presents the boxplot for CO₂ emission intensity. In contrast to consumption, the distribution is markedly asymmetric. The median is approximately 100 g/kWh, while the interquartile range spans roughly 50 to 160 g/kWh. Many high-value outliers are visible above the upper whisker, with several observations exceeding 400 g/kWh, corresponding to episodes of low renewable generation combined with high reliance on conventional thermal generation or high-carbon imports.

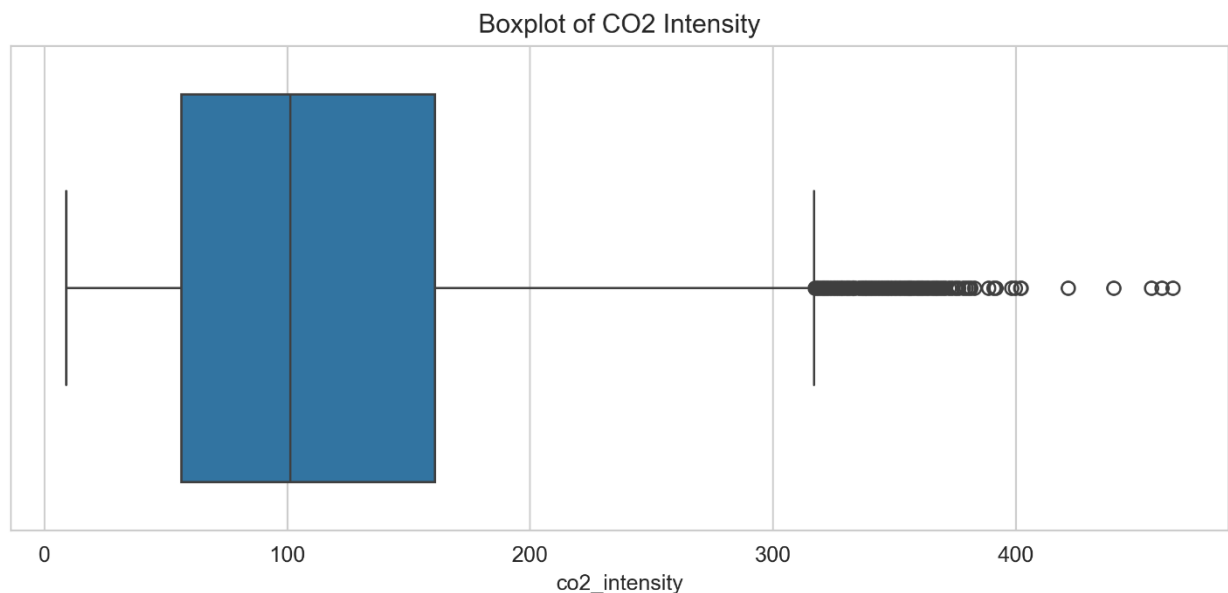


Figure 3.11. Boxplot of CO₂ emission intensity. The distribution is right-skewed with a low median and a substantial number of high-value outliers extending well beyond the upper whisker, reflecting episodic spikes in carbon intensity during periods of reduced renewable generation.

3.7 Summary

This chapter has described the data sources, preprocessing procedures, feature engineering, exploratory analysis for forecasting hourly electricity consumption and CO₂ emission intensity in Denmark's DK1 bidding zone. Three datasets - Energinet production and consumption settlement data, Energinet CO₂ emission intensity data, and DMI weather observations from Station 06072 - were integrated into a unified hourly dataset covering January 2022 through December 2024. Preprocessing steps included temporal aggregation (5-minute and 10-minute data to hourly resolution), timestamp standardization to UTC, and geographic filtering to DK1.

Feature engineering created aggregated renewable and conventional generation variables, cross-border flow summaries, time-based calendar features (hour, day of week, month, weekend indicator, and season), lagged versions of the target variables at 1-hour, 24-hour, and 168-hour intervals, and lag-1 versions of production and exchange variables to capture recent energy mix information without introducing data leakage.

Exploratory analysis revealed clear daily and weekly seasonality in electricity consumption, and a more complex and volatile temporal pattern for CO₂ intensity. Autocorrelation analysis confirmed strong autocorrelation at lags 1, 24, and 168 for consumption (0.965, 0.822, and 0.815 respectively), empirically justifying the lag selection. For CO₂ intensity, while lag-1 autocorrelation is high (0.947), autocorrelation decays substantially at lag 24 (0.510) and lag 168 (0.227), indicating that CO₂ intensity requires additional explanatory variables beyond lagged values alone. Analysis of predictor-target relationships confirmed a nonlinear U-shaped relationship between temperature and consumption and a strong inverse relationship between wind generation and CO₂ intensity, both motivating the use of flexible machine learning models.

The final model-ready dataset contains 26,135 complete hourly observations across 28 columns. The modeling framework, data split strategy, and evaluation approach are described in Chapter 4.

CHAPTER 4: MODELING AND METHODOLOGY

4.1 Introduction

This chapter describes the modeling framework used to forecast hourly electricity consumption and CO₂ emission intensity in the DK1 bidding zone. The chapter is organized as follows. Section 4.2 defines the forecasting problem formally. Section 4.3 describes the train-validation-test split strategy. Section 4.4 presents the two modeling specifications used throughout the analysis. Section 4.5 describes the three model families: naïve persistence, linear regression, and gradient boosting. Section 4.6 covers hyperparameter tuning. Section 4.7 describes the evaluation metrics. Section 4.8 describes the feature importance and SHAP methodology.

4.2 Forecasting Problem Definition

The forecasting task is formulated as a one-hour-ahead point forecasting problem. For each hourly observation at time t , the objective is to predict the target value at hour $t+1$ using only information that would realistically be available at or before the forecast time. Formally, the forecasting problem is defined as:

$$\hat{y}(t+1) = f(X_t)$$

where $\hat{y}(t+1)$ denotes the predicted target value for the next hour and X_t represents the feature vector constructed from information available up to time t .

This forecasting setup is applied independently to two target variables:

- **GrossConsumptionMWh**: hourly electricity consumption in DK1, measured in megawatt-hours (MWh)
- **co2_intensity**: hourly CO₂ emission intensity in DK1, measured in grams of CO₂ per kilowatt-hour (gCO₂/kWh)

A one-hour-ahead forecasting horizon was selected because it is operationally relevant for short-term balancing and reserve scheduling in electricity systems. The forecasting setup also determines which variables are admissible as model inputs without introducing data leakage.

To reduce the risk of leakage, contemporaneous electricity system variables - including production mix, renewable generation, imports, exports, and net exchange - were not used directly as predictors. Instead, these variables were incorporated only through lagged versions, primarily lag-1 features representing the most recently observed system state. Similarly, lagged values of electricity consumption and CO₂ intensity were included to capture short-term persistence, daily seasonality, and weekly temporal patterns.

Calendar-based variables such as hour of day, day of week, month, weekend indicator, and season were retained because they are known in advance and therefore do not introduce look-ahead bias.

Weather variables - temperature, wind speed, and solar radiation - were included under the assumption that short-term weather forecasts are available for the forecast horizon, which is a standard assumption in short-horizon energy forecasting (Hong & Fan, 2016). In the empirical implementation, observed weather measurements were used as proxies for these forecasted weather inputs. Consequently, the reported forecasting performance should be interpreted as an idealized or upper-bound forecasting scenario rather than a fully operational real-time forecasting deployment using actual weather forecasts.

This distinction is important because the models do not use future target values or contemporaneous production and exchange variables, but they do assume access to accurate short-term weather information. Therefore, the empirical setup should be interpreted as a leakage-controlled one-hour-ahead forecasting experiment designed to evaluate the predictive value of the selected feature set under controlled conditions.

4.3 Train-Validation-Test Split

A strict temporal split was applied to preserve the ordering of observations and prevent any form of future information leaking into model training or evaluation. The dataset was divided into three consecutive non-overlapping periods:

- **Training set**: January 2022 - September 2023 (~15,143 observations after removing rows with missing lag values)
- **Validation set**: October 2023 - December 2023 (2,208 observations)
- **Test set**: January 2024 - December 2024 (8,784 observations)

The validation set was used exclusively for hyperparameter tuning. The test set was held out entirely until final model evaluation and was never used to make any modeling decisions. This structure ensures that reported test performance reflects genuine out of sample accuracy (Hyndman & Athanasopoulos, 2021).

After hyperparameter selection on the validation set, the final model was retrained on the combined training and validation data before evaluation on the test set. This is a standard practice that makes full use of available data once the model configuration is fixed (Hastie et al., 2009).

Random splits and k-fold cross-validation were deliberately avoided. In time-series forecasting, random partitioning allows the model to observe future data during training, producing overoptimistic performance estimates that do not reflect real operational conditions (Bergmeir & Benítez, 2012).

4.4 Modeling Specifications

Two complementary modeling specifications were used throughout the analysis, serving distinct purposes.

Specification 1: Predictive modeling setup (with lag-1): This specification includes the lag-1 target variable (cons_lag_1 for consumption, co2_lag_1 for CO₂ intensity) as a predictor alongside all other features. It is designed to maximize short-term forecasting accuracy and represents the primary framework for evaluating predictive performance. This setup directly addresses RQ1 and RQ2.

Specification 2: Diagnostic modeling setup (without lag-1): This specification excludes the lag-1 target variable. Its purpose is not to produce a more accurate forecast removing the strongest predictor inevitably worsens accuracy but to examine how models utilize weather variables, lagged system variables, and calendar features when the dominant autoregressive signal is absent. This setup directly addresses RQ3, because when lag-1 is included, its overwhelming dominance in feature importance obscures the contribution of all other predictors (Lundberg & Lee, 2017). The no-lag-1 specification should therefore be interpreted as a diagnostic sensitivity analysis, not as an alternative operational forecasting model. Any feature importance results from this setup reflect the model's behavior in the absence of persistence and should not be interpreted as the primary forecasting specification.

4.5 Models

4.5.1 Naïve Persistence Baseline

Two naïve persistence baselines were evaluated to provide a simple reference benchmark on meaningful model performance.

The **lag-1 persistence model** forecasts the target variable at hour $t+1$ as equal to the observed value at hour t :

$$\hat{y}(t+1) = y(t)$$

For a one-hour-ahead forecasting task, lag-1 persistence is the natural and most demanding benchmark. Electricity consumption and CO₂ intensity both exhibit strong short-term autocorrelation, lag-1 values of 0.965 and 0.947 respectively, as documented in Chapter 3 meaning the previous hour's observation is already an informative predictor. Any model that cannot improve upon this benchmark is not adding meaningful value at this horizon.

The **lag-24 persistence model** forecasts using the same hour from the previous day:

$$\hat{y}(t+1) = y(t-23)$$

This baseline captures daily seasonality but is less informative than lag-1 at a one-hour horizon. It is included as a secondary reference point to illustrate how much worse a day-ahead naive forecast performs compared to short-term persistence.

4.5.2 Linear Regression

A multivariate ordinary least squares (OLS) linear regression model was fitted for each target variable. The model takes the form:

$$\hat{y}(t+1) = \beta_0 + \beta_1 x_1(t) + \beta_2 x_2(t) + \dots + \beta_p x_p(t) + \varepsilon(t)$$

where $x_1(t)$, ..., $x_p(t)$ are the predictor variables available at time t , β_0 is the intercept, β_1 , ..., β_p are the estimated coefficients, and $\varepsilon(t)$ is the residual error term.

The predictor set includes lagged target variables, weather variables (temperature, wind speed, solar radiation), calendar features (hour of day, day of week, month, weekend indicator, seasonal dummies), and lagged production and exchange variables (lag-1 values of wind, solar, renewables, conventional generation, total production, imports, exports, and net exchange).

Linear regression serves as an interpretable intermediate baseline between naïve persistence and gradient boosting. It captures the main additive effects of each predictor but assumes that all relationships are linear and that there are no interaction effects between variables. This is a meaningful limitation in the DK1 context, where the relationship between temperature and consumption is nonlinear (U-shaped) and where wind generation interacts with cross-border flows to determine CO₂ intensity in complex ways (Leerbeck et al., 2020). Comparing linear regression against gradient boosting therefore directly quantifies the value of capturing these nonlinearities, which is central to RQ2.

4.5.3 Gradient Boosting

Gradient boosting is an ensemble learning method that constructs a strong predictive model by sequentially adding weak learners - typically shallow decision trees - each trained to correct the residual errors of the previous ensemble (Friedman, 2001). The final prediction is an additive combination of all trees:

$$F_M(x) = \sum_m \gamma_m h_m(x)$$

where $h_m(x)$ is the m -th decision tree, γ_m is its weight, and M is the total number of trees. Each tree is fitted to the negative gradient of the loss function with respect to the current ensemble's predictions, a process known as functional gradient descent (Mason et al., 1999).

The key advantages of gradient boosting for this forecasting task are threefold. First, it can capture nonlinear relationships between predictors and targets automatically, without requiring explicit specification of interaction or polynomial terms. Second, tree-based gradient boosting can flexibly model nonlinear patterns in irregular or skewed data, although sensitivity to outliers depends on the loss function and hyperparameters. Third, it provides built-in feature importance measures that can be used to interpret which predictors drive the forecasts (Hastie et al., 2009).

The implementation uses scikit-learn's GradientBoostingRegressor (Pedregosa et al., 2011). Although the original research proposal referred to XGBoost (Chen & Guestrin, 2016), GradientBoostingRegressor was used in the final implementation for its transparency and direct compatibility with the SHAP analysis pipeline. Both belong to the same gradient boosting family and share the same underlying algorithmic principles. The distinction is made explicit here to ensure full consistency between the reported methods and the actual implementation.

The same predictor set used in linear regression is applied to the gradient boosting models, ensuring that any performance differences are attributable to the modeling approach rather than differences in input features.

4.6 Hyperparameter Tuning

Gradient boosting performance is sensitive to the choice of hyperparameters. Three key parameters were tuned:

- **n_estimators**: the number of sequential trees in the ensemble
- **max_depth**: the maximum depth of each individual tree
- **learning_rate**: the shrinkage factor applied to each tree's contribution

A grid search was conducted over the following values:

- $n_estimators \in \{100, 200\}$
- $max_depth \in \{3, 5\}$
- $learning_rate \in \{0.05, 0.1\}$

This produces eight candidate configurations per target variable. Each configuration was evaluated using the validation set MAE, with temporal ordering strictly preserved. The configuration minimizing validation MAE was selected as the final model.

For **electricity consumption**, the best configuration was $n_estimators=100$, $max_depth=5$, $learning_rate=0.05$.

For **CO₂ intensity**, the best configuration was $n_estimators=200$, $max_depth=3$, $learning_rate=0.05$.

After hyperparameter selection, each final model was retrained on the combined training and validation data to make maximum use of available observations before test set evaluation. All tuning decisions were made without any access to test set observations.

4.7 Evaluation Metrics

Model performance is evaluated using two standard regression metrics computed on the held out 2024 test set.

Mean Absolute Error (MAE):

$$MAE = (1/n) \sum_i |y_i - \hat{y}_i|$$

MAE measures the average absolute deviation between predicted and observed values. It is expressed in the same units as the target variable MWh for consumption, gCO₂/kWh for CO₂ intensity and is directly interpretable as the typical forecast error. MAE is less sensitive to large individual errors than squared-error metrics, making it appropriate for comparing overall model accuracy (Hyndman & Koehler, 2006).

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{(1/n) \sum_i (y_i - \hat{y}_i)^2}$$

RMSE penalizes large errors more heavily than MAE by squaring the residuals before averaging. This makes it more sensitive to large forecast deviations - which are operationally important in electricity systems, where large errors during peak demand or high-CO₂ events have disproportionate consequences (Hong & Fan, 2016).

Both metrics are reported for each model and each target variable. To facilitate comparison across models, improvements over the lag-1 persistence baseline are also expressed as percentage reductions in MAE:

$$\% \text{ improvement} = ((\text{MAE_baseline} - \text{MAE_model}) / \text{MAE_baseline}) \times 100$$

Negative values indicate that the model performs worse than the baseline.

The two targets are kept strictly separate throughout the evaluation. Electricity consumption and CO₂ intensity are measured on entirely different scales, and comparing their raw error magnitudes directly would be misleading.

4.8 Feature Importance and SHAP Analysis

To address RQ3 which input variables contribute most to predictive performance there are two complementary feature importance methods which are applied to the gradient boosting models.

4.8.1 Gain-Based Feature Importance

Gradient boosting models provide a built-in measure of feature importance based on the total reduction in the loss function attributable to each feature across all splits in all trees, referred to as the gain (Friedman, 2001). Features that consistently reduce prediction error when used for splitting receive higher importance scores. Gain-based importance is computationally efficient and provides a quick overview of predictor relevance.

However, gain-based importance has known limitations. It is biased toward features with higher cardinality and does not reveal the direction of effects or interactions between features (Strobl et al., 2007). For this reason, it is complemented by SHAP analysis.

4.8.2 SHAP Values

SHAP (SHapley Additive exPlanations), introduced by (Lundberg & Lee, 2017), provides a theoretically grounded approach to model interpretability based on Shapley values from cooperative game theory (Shapley, 1953). For each prediction, SHAP assigns a contribution value ϕ_i to each feature i such that:

$$f(x) = \phi_0 + \sum_i \phi_i$$

where $f(x)$ is the model prediction and ϕ_0 is the base value (average model output). The SHAP values satisfy three desirable properties: local accuracy (contributions sum to the prediction), missingness (absent features receive zero contribution), and consistency (a feature's contribution does not decrease if its effect increases) (Lundberg & Lee, 2017).

SHAP values are computed using the shap Python library. Global feature importance is assessed by averaging absolute SHAP values across all test set observations. An important interpretive caution applies throughout: SHAP values describe how the fitted model uses its inputs, not why the underlying system behaves as it does. In particular, they do not imply causal relationships between predictors and targets (Molnar, 2022). This distinction is maintained throughout the results chapter.

4.9 Summary

This chapter has defined the forecasting problem, described the temporal data split, and presented the three model families used in this study: naïve persistence, linear regression, and gradient boosting. Two modeling specifications are used throughout: a full predictive setup that includes the lag-1 target variable and is optimized for forecast accuracy, and a diagnostic setup that excludes lag-1 to reveal the contribution of exogenous predictors. Model performance is evaluated on the held-out 2024 test set using MAE and RMSE, with improvements expressed relative to the lag-1 persistence baseline. Feature importance is assessed through gain-based measures and SHAP values, interpreted as model diagnostics rather than causal evidence. The results of this evaluation are presented in Chapter 5.

CHAPTER 5: RESULTS

5.1 Introduction

This chapter presents the forecasting results for electricity consumption and CO₂ emission intensity across the 2024 test period. The structure follows the two modeling specifications defined in Chapter 4. Section 5.2 covers electricity consumption forecasting, first the full predictive setup with lag-1, then the diagnostic setup without it. Section 5.3 covers CO₂ intensity forecasting in the same structure. Each section includes aggregate performance metrics, feature importance and SHAP analysis, and conditional error analysis across hours of the day, seasons, and renewable generation levels. Section 5.4 summarizes the key findings across both targets.

The two targets are kept strictly separate throughout. Their raw error magnitudes are not comparable; consumption is measured in MWh while CO₂ intensity is measured in gCO₂/kWh. So, all cross-model comparisons within each target use percentage improvements relative to the lag-1 persistence baseline. Sections 5.2.2 and 5.3.2 report aggregate model performance and directly address RQ1 and RQ2. Sections 5.2.4 and 5.3.4 present the diagnostic no-lag-1 analysis, which is the primary basis for addressing RQ3.

5.2 Electricity Consumption Forecasting

5.2.1 Baseline Performance

Two naïve persistence baselines were evaluated before fitting any statistical model. The lag-1 baseline carries the previous hour's consumption forward as the forecast. The lag-24 baseline uses the same hour from the previous day.

On the 2024 test set, the lag-1 baseline achieves a MAE of 102.63 MWh and RMSE of 133.46 MWh. The lag-24 baseline performs substantially worse at a MAE of 206.90 MWh and RMSE of 285.46 MWh - a deterioration of 102% over the lag-1 baseline.

This gap reflects the fundamental structure of the forecasting task. At a one-hour horizon, electricity consumption changes gradually and the previous hour is far more informative than the same hour yesterday. This is consistent with the ACF analysis in Chapter 3, where lag-1 autocorrelation for consumption was 0.965 compared to 0.822 at lag-24. The lag-1 baseline is therefore the meaningful benchmark against which all models are evaluated.

5.2.2 Full Model Performance

Table 5.1 summarizes the complete test set results for electricity consumption across all models.

Table 5.1. Electricity consumption forecasting results on the 2024 test set.

Model	Test MAE(MWh)	Test RMSE(MWh)	% vs lag-1 MAE
Lag-1 persistence	102.63	133.46	-
Lag-24 persistence	206.90	285.46	-101.6%
Linear regression	85.33	111.62	+16.9%
Gradient boosting (initial)	68.97	96.92	+32.8%
Gradient boosting (tuned)	65.13	92.49	+36.5%
Gradient boosting (no lag-1)	107.54	145.85	-4.8%

Linear regression reduces the MAE to 85.33 MWh, a 16.9% improvement over lag-1. Incorporating weather, calendar, and lagged production variables into a linear framework already adds predictive value beyond simply repeating the last observation.

Gradient boosting improves further. The initial model achieves a MAE of 68.97 MWh. After grid search tuning on the validation set, the best configuration - `n_estimators=100`, `max_depth=5`, `learning_rate=0.05`, achieves a test MAE of 65.13 MWh and RMSE of 92.49 MWh. This represents a 36.5% reduction over the lag-1 baseline and a 23.7% improvement over linear regression. The additional gain from gradient boosting over linear regression suggests that nonlinear patterns and interactions provide useful predictive information. This is consistent with the U-shaped temperature-consumption relationship and time-dependent demand patterns identified in Chapter 3, although the full-model feature importance remains dominated by `cons_lag_1`.

Practical significance: The average electricity consumption during the test period is 2,748.6 MWh. The lag-1 baseline MAE of 102.63 MWh represents 3.73% of average consumption. The tuned gradient boosting model reduces this to 65.13 MWh - 2.37% of average consumption. In absolute terms, this is a reduction in typical forecast error of approximately 37.5 MWh per hour - this is a substantial reduction in average hourly forecast error. Its direct operational value would require evaluation against reserve requirements, market rules, and forecast-error costs.

Average consumption during the 2024 test period (2,748.6 MWh) is higher than the full dataset mean reported in Chapter 3 (approximately 2,630 MWh), reflecting year-to-year variation in demand across the study period.

5.2.3 Feature Importance and SHAP: Full Model

An important interpretive caution applies before reading these results: feature importance from the full model should not be taken as the answer to RQ3. When `cons_lag_1` is included, its dominance reflects the autocorrelation structure of the series rather than the contribution of exogenous variables to forecasting electricity consumption. The diagnostic no-lag-1 setup in Section 5.2.4 is the correct basis for addressing RQ3. With that caveat stated, the gain-based scores reveal a striking result: `cons_lag_1` accounts for 95.1% of total importance. The next most important feature is hour of day at 3.1%, followed by `cons_lag_168` at 0.7% and `cons_lag_24` at 0.3%. All remaining features weather variables, production lags, exchange lags - contribute less than 0.3% combined.

The SHAP summary plot (Figure 5.1) confirms this picture. The dot cloud for `cons_lag_1` spans a far wider range of SHAP values than any other feature, indicating that it drives the vast majority of the

model's predictions. The relationship is monotonic - higher lag-1 consumption leads to higher predicted consumption - which is consistent with the strong autocorrelation structure of the series.

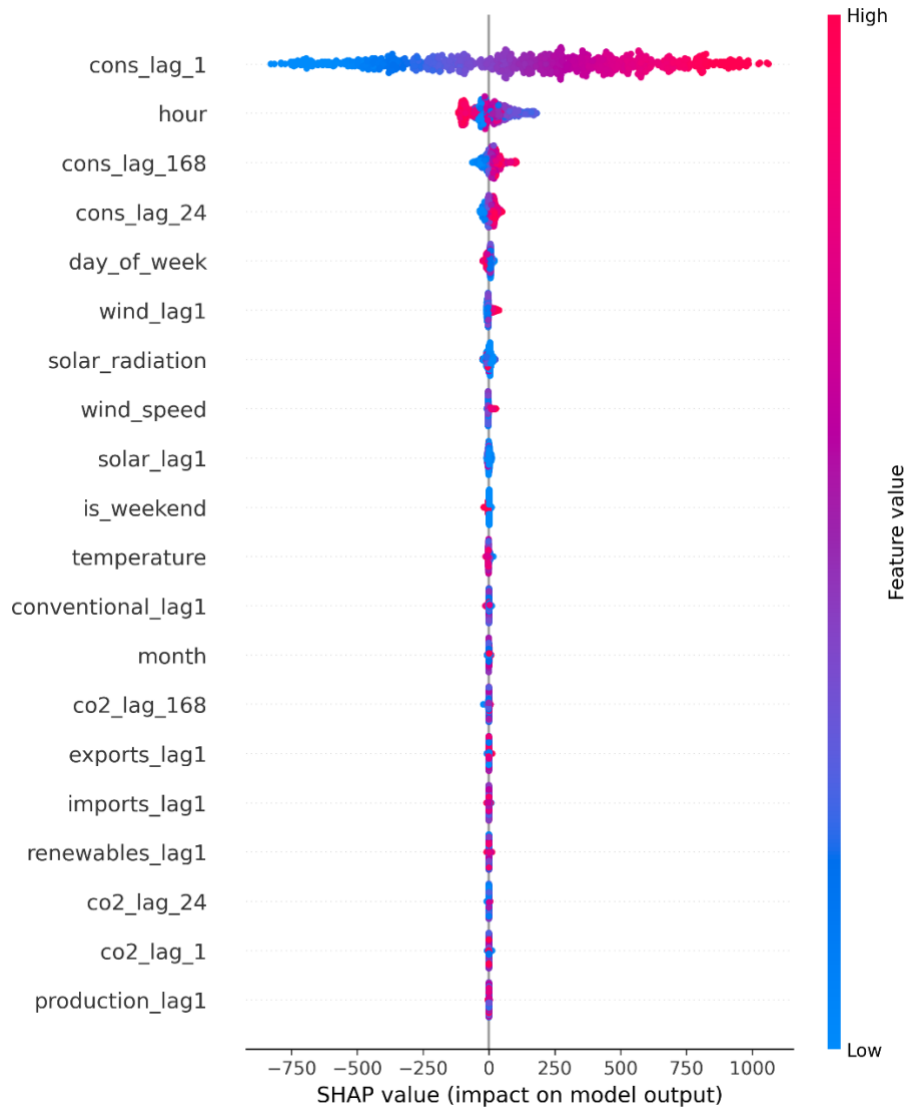


Figure 5.1. SHAP summary plot for the tuned gradient boosting model predicting electricity consumption (full model with `cons_lag_1`). Each dot represents one test set observation. The horizontal position shows the SHAP value (contribution to the prediction) and the color indicates the feature value. `cons_lag_1` dominates all other features, accounting for 95.1% of total gain-based importance.

These results should be read as a diagnostic of model behavior rather than as evidence about what drives electricity consumption in DK1. The dominance of `cons_lag_1` reflects the strong short-term autocorrelation of the series. When the previous observation is available, the gradient boosting model essentially learns a refined version of lag-1 persistence with a small correction from hour of day. All other features are effectively marginal at this horizon.

5.2.4 Diagnostic Setup: Sensitivity Analysis

When `cons_lag_1` is removed from the feature set, test MAE rises to 107.54 MWh and RMSE to 145.85 MWh, slightly worse than the lag-1 persistence baseline. This confirms that removing the strongest predictor costs more than the remaining features can recover. The sensitivity analysis should not be

interpreted as a competitive forecasting approach but as a diagnostic tool for understanding which variables carry predictive weight when the autoregressive shortcut is unavailable.

The SHAP summary plot from this specification (Figure 5.2) reveals a very different picture from the full model. Lagged net exchange and lagged total production emerge as the most important predictors, followed by `cons_lag_24`, `wind_lag1` and `cons_lag_168`. Weather variables - solar radiation, wind speed, temperature - contribute more noticeably than in the full model, though still less than system state variables.

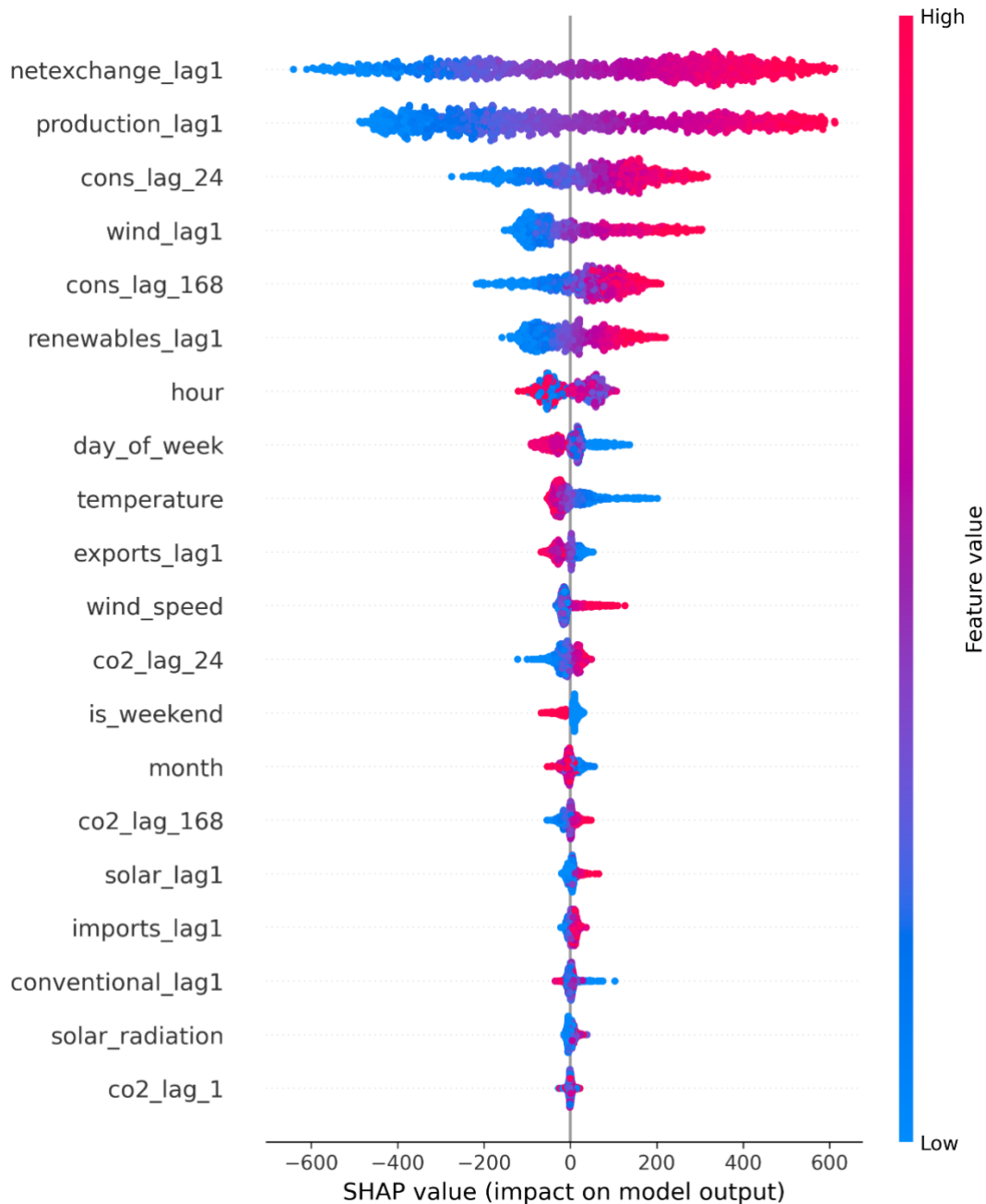


Figure 5.2. SHAP summary plot for the gradient boosting model predicting electricity consumption without `cons_lag_1` (diagnostic setup). With the dominant autoregressive predictor removed, lagged net exchange and lagged total production emerge as the most important predictors, followed by the 24-hour and 168-hour consumption lags.

This pattern is physically coherent for DK1. When the previous hour's consumption is unavailable, the model draws on information about the system's current operating state - how much electricity is being generated and how much is crossing the interconnectors. These are informative correlates of consumption in a zone where generation can substantially exceed or fall short of local demand. The longer lags (cons_lag_24, cons_lag_168) then add the daily and weekly routine around that system state.

5.2.5 Conditional Performance

The aggregate MAE of 65.13 MWh conceals meaningful variation across different system conditions.

By hour of day: Figure 5.3 shows forecast errors by hour. Errors are lowest during the late evening and overnight hours - hour 20 sees a MAE of 40.84 MWh and hour 22 sees 44.13 MWh. Errors peak during the mid-morning transition at hour 10 (92.06 MWh) and the evening consumption surge at hour 17 (89.11 MWh). These are precisely the periods identified in Chapter 3 as having the steepest and most variable consumption transitions - the morning rise from overnight lows and the evening surge as residential demand increases. The model struggles most where consumption is changing fastest.

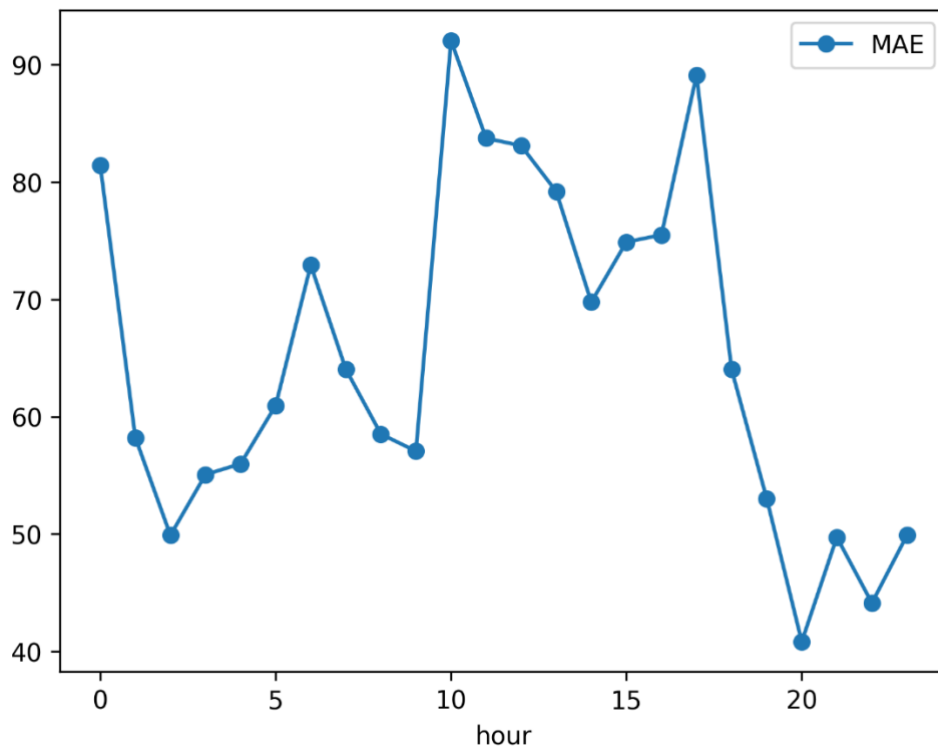


Figure 5.3. Mean absolute error (MAE) of the tuned gradient boosting model for electricity consumption by hour of day across the 2024 test set. Errors are lowest during overnight hours and peak at hour 10 (92.06 MWh) and hour 17 (89.11 MWh), corresponding to the morning and evening consumption transition periods.

By season: Summer is the easiest period to forecast with a MAE of 55.14 MWh, while winter (70.06 MWh) and spring (71.11 MWh) are harder. Summer demand is lower in absolute terms and more stable, while winter demand is both higher and more sensitive to temperature - the same weather-driven nonlinearity that motivated gradient boosting in the first place. Autumn sits in between at 64.27 MWh.

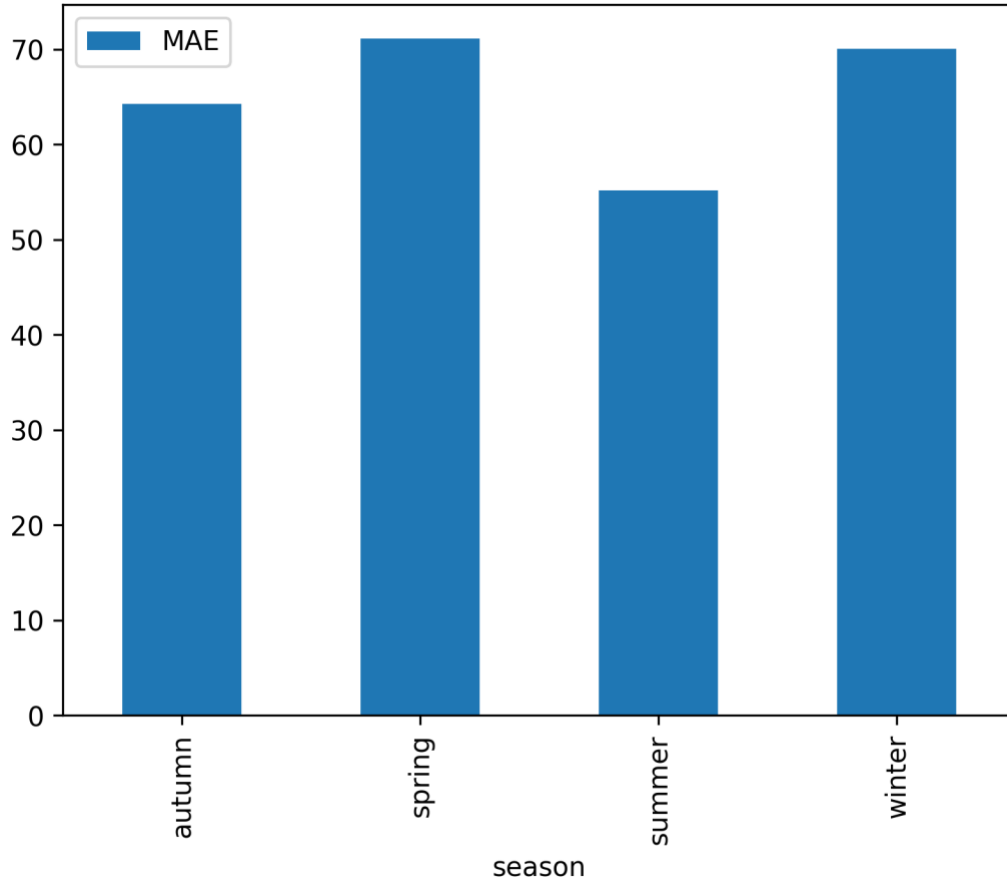


Figure 5.4. Mean absolute error (MAE) of the tuned gradient boosting model for electricity consumption by season across the 2024 test set. Summer is the easiest season to forecast (55.14 MWh) while winter and spring are harder (70.06 and 71.11 MWh respectively), reflecting greater temperature-driven demand variability in colder months.

By renewable generation level: Conditional error analysis groups forecasts by terciles of lagged renewable generation (`renewables_lag1`). Errors increase monotonically with renewable generation: low-renewable periods see a MAE of 47.92 MWh, medium-renewable periods 65.62 MWh, and high-renewable periods 81.85 MWh. High renewable generation in DK1 is associated with high wind output, which introduces more volatile system conditions overall. During these periods, cross-border flows are more active and the relationship between consecutive hours becomes less predictable, making the lag-1 signal less reliable as an anchor for the forecast.

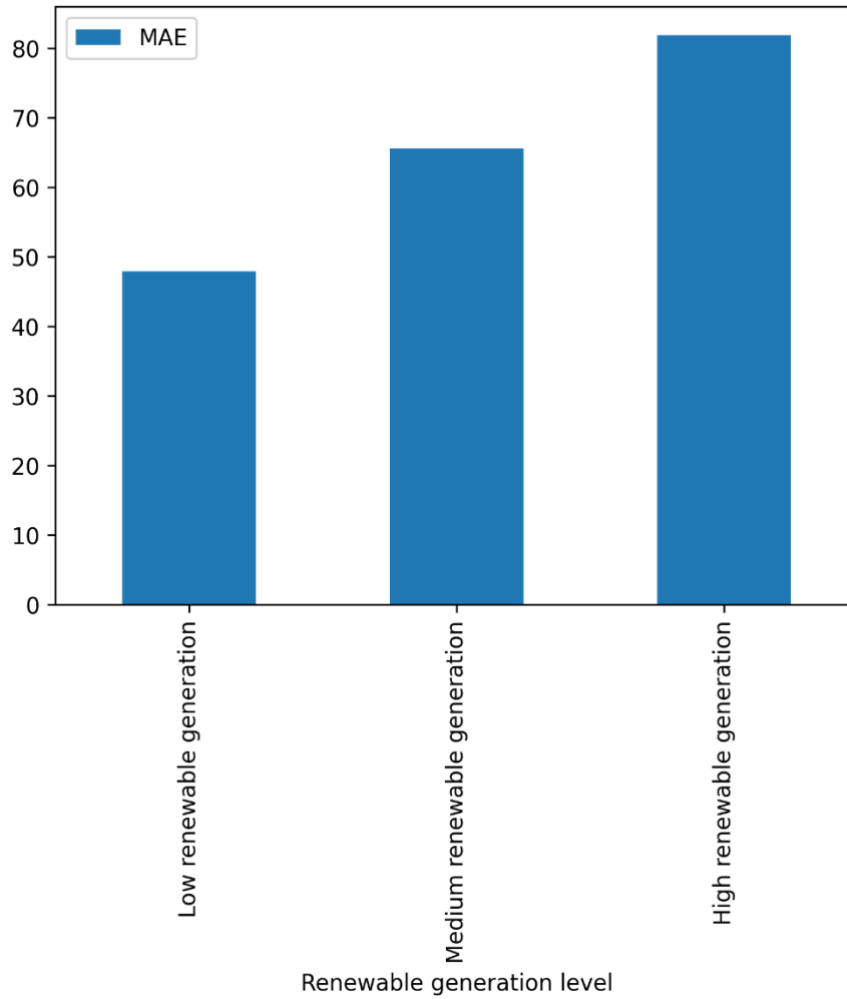


Figure 5.5. Mean absolute error (MAE) of the tuned gradient boosting model for electricity consumption by lagged renewable generation level across the 2024 test set. Errors increase monotonically with renewable generation - low (47.92 MWh), medium (65.62 MWh), and high (81.85 MWh) - reflecting greater system volatility during high-wind periods.

5.3 CO₂ Emission Intensity Forecasting

5.3.1 Baseline Performance

The lag-1 baseline achieves a test MAE of 15.52 gCO₂/kWh and RMSE of 24.06 gCO₂/kWh. The lag-24 baseline performs dramatically worse - MAE of 52.47 gCO₂/kWh and RMSE of 68.65 gCO₂/kWh - a deterioration of 238% relative to lag-1.

This contrast is larger than for consumption and reflects the ACF structure documented in Chapter 3. CO₂ intensity has a lag-1 autocorrelation of 0.947 but only 0.510 at lag-24 and 0.227 at lag-168. Unlike consumption, CO₂ intensity does not follow a predictable daily or weekly cycle - it is driven by renewable generation availability, which follows weather patterns rather than the clock. The same hour yesterday tells you very little about today's carbon intensity. The previous hour tells you a great deal.

5.3.2 Full Model Performance

Table 5.2 summarizes the complete test set results for CO₂ emission intensity.

Table 5.2. CO₂ emission intensity forecasting results on the 2024 test set.

Model	Test MAE (gCO ₂ /kWh)	Test RMSE (gCO ₂ /kWh)	% vs lag-1 MAE
Lag-1 persistence	15.52	24.06	-
Lag-24 persistence	52.47	68.65	-238.2%
Linear regression	16.01	23.31	-3.2%
Gradient boosting (initial)	15.75	23.21	-1.5%
Gradient boosting (tuned)	15.70	23.18	-1.2%
Gradient boosting (no co2_lag_1)	36.74	45.76	-136.7%

The result here is qualitatively different from consumption: neither linear regression nor gradient boosting improves over the lag-1 baseline in MAE. Linear regression is 3.2% worse. The tuned gradient boosting model, best configuration `n_estimators=200`, `max_depth=3`, `learning_rate=0.05`, achieves a MAE of 15.70 gCO₂/kWh, which is 1.2% worse than the lag-1 baseline.

This is a substantive finding in its own right. At the one-hour horizon and with the information set available here, lag-1 persistence is the best achievable forecast in MAE terms among the tested models. CO₂ intensity appears strongly associated with the current generation mix and exchange situation, especially wind generation in DK1, which changes on atmospheric timescales that lagged system variables and historical weather observations cannot reliably anticipate. The lag-1 value efficiently encodes the current renewable penetration state, and no additional features can consistently improve upon it within a one-hour window.

The RMSE tells a marginally different story. The tuned gradient boosting model achieves 23.18 gCO₂/kWh compared to the lag-1 baseline's 24.06, a 3.7% reduction. Since RMSE penalizes large errors more heavily, this suggests gradient boosting handles the high-intensity spikes which occur during low-wind, high-import periods, slightly better than persistence. But the practical significance of a 0.88 gCO₂/kWh improvement in RMSE is modest. The main conclusion is, for CO₂ intensity at this horizon, persistence is the appropriate forecasting benchmark, and trained models do not meaningfully improve upon it.

Practical significance: The average CO₂ intensity during the test period is 103.02 gCO₂/kWh. The lag-1 baseline MAE of 15.52 gCO₂/kWh represents 15.06% of average intensity. The tuned gradient boosting model achieves 15.70 gCO₂/kWh, 15.24% of average intensity. The additional complexity of gradient boosting does not translate into a meaningful reduction in relative forecast error for this target.

Similarly, the average CO₂ intensity during the 2024 test period (103.02 gCO₂/kWh) is lower than the full dataset mean of approximately 114.8 gCO₂/kWh reported in Chapter 3, reflecting the continued growth of renewable generation in DK1 over the study period - 2024 was a cleaner year on average than 2022 and 2023.

5.3.3 Feature Importance and SHAP: Full Model

The feature importance results are even more concentrated than for consumption. `co2_lag_1` accounts for approximately 98% of total gain-based importance. All remaining features `exports_lag1`, `conventional_lag1`, `co2_lag_24`, `hour`, `wind_lag1` each contribute less than 0.2%.

The SHAP summary plot (Figure 5.6) confirms this. The dot cloud for `co2_lag_1` dominates the plot by a wide margin. The relationship is clearly monotonic, higher previous-hour intensity predicts higher next-hour intensity reflecting the strong short-term persistence of the series.

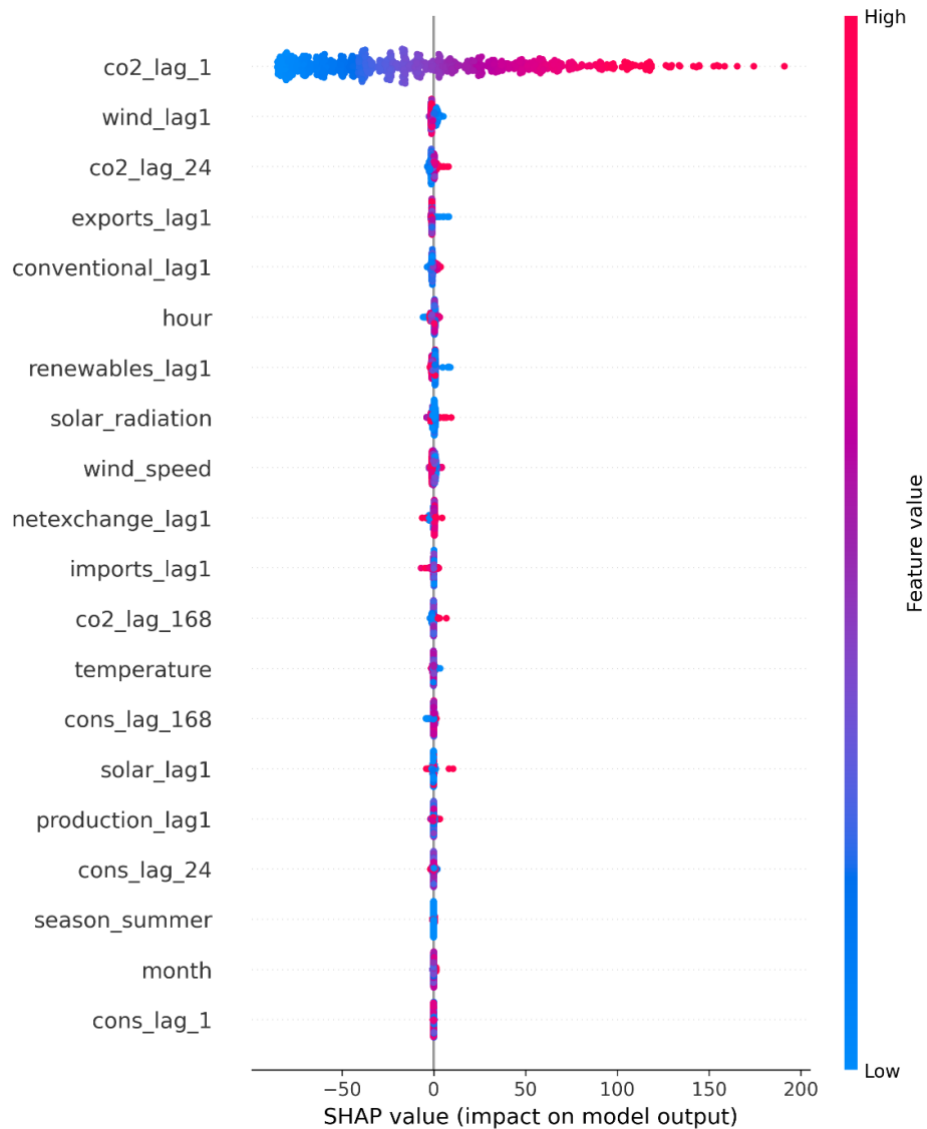


Figure 5.6. SHAP summary plot for the tuned gradient boosting model predicting CO₂ emission intensity (full model with `co2_lag_1`). `co2_lag_1` accounts for approximately 98% of total gain-based importance, indicating that the model relies almost entirely on the previous hour's intensity value. All other features contribute marginally.

As with consumption, these results describe how the model uses its inputs and should not be interpreted as evidence about the causal structure of the CO₂ system. The near-complete dominance of `co2_lag_1` means the full model provides almost no information about which exogenous variables matter. That question is addressed by the diagnostic setup.

5.3.4 Diagnostic Setup: Sensitivity Analysis

Removing `co2_lag_1` results in a test MAE of 36.74 gCO₂/kWh and RMSE of 45.76 gCO₂/kWh, more than double the full model's MAE. The deterioration is larger than the corresponding drop for consumption, consistent with the even stronger role of persistence in the CO₂ series.

Despite this, the sensitivity analysis still outperforms the lag-24 baseline substantially (MAE 36.74 vs 52.47 gCO₂/kWh), confirming that the lagged system variables do carry genuine predictive information. It is simply that this information is largely already captured by the lag-1 CO₂ value when it is available.

The SHAP summary plot from this specification (Figure 5.7) reveals a physically meaningful picture. `conventional_lag1` and `renewables_lag1` emerge as the dominant predictors, followed by `imports_lag1` and `co2_lag_24`. `wind_lag1` also appears among the important predictors. Temperature and solar radiation contribute less, reflecting the secondary role of solar generation in DK1's predominantly wind-driven mix.

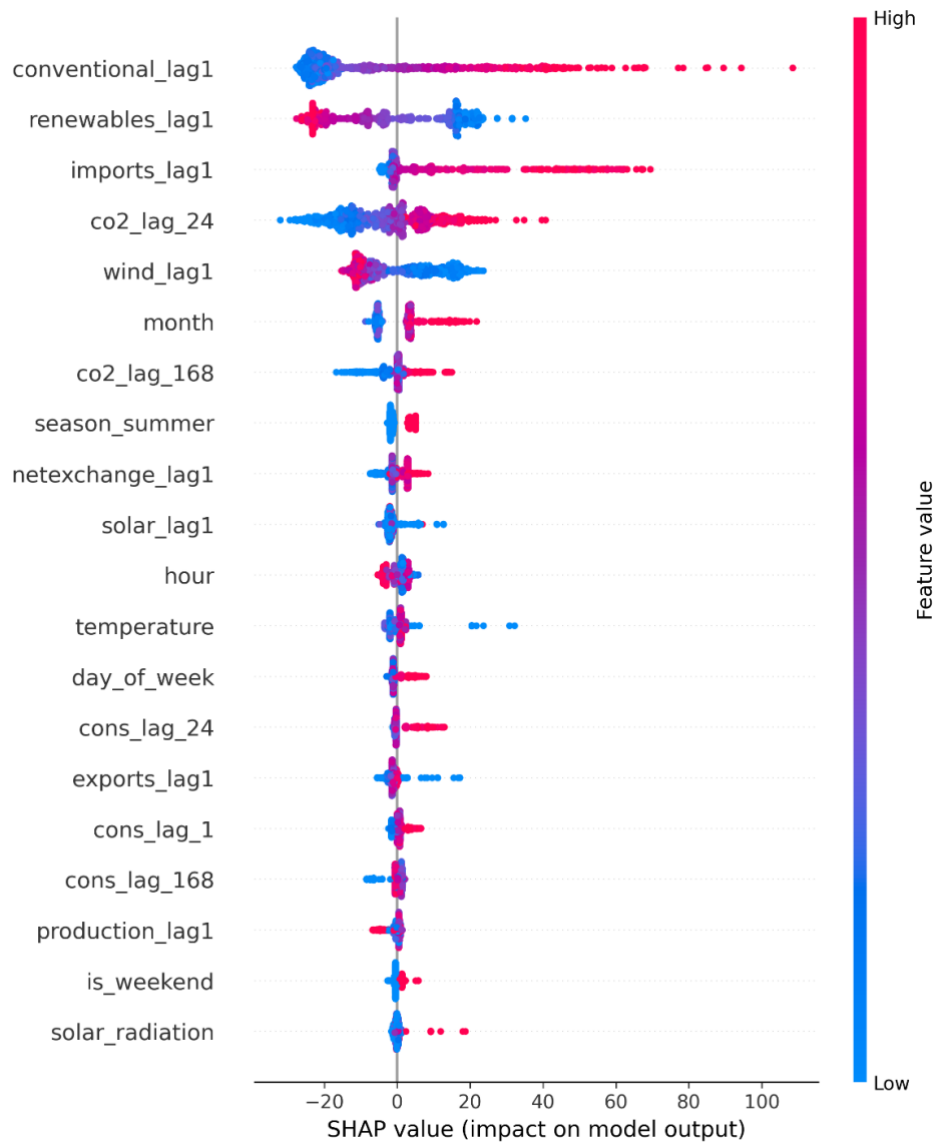


Figure 5.7. SHAP summary plot for the gradient boosting model predicting CO₂ emission intensity without `co2_lag_1` (diagnostic setup). `conventional_lag1` and `renewables_lag1` emerge as the dominant predictors.

This result connects directly to the generation mix dynamics documented in Chapter 3: when conventional generation is high, CO₂ intensity tends to be high; when renewables are high, it tends to be low. The sensitivity analysis is identifying these same system-state relationships from lagged data. `Conventional_lag1` appearing as an important predictor is also interpretable: when conventional generation was high in the previous hour, it signals a low-renewable system state likely to persist into the next hour.

5.3.5 Conditional Performance

The conditional analysis for CO₂ is based on the lag-1 persistence model, which provides the best overall forecasting performance for this target.

By hour of day: Figure 5.8 shows that errors are lowest during overnight hours. Hour 3 sees a MAE of 11.59 gCO₂/kWh and hour 2 sees 11.84 gCO₂/kWh. Errors peak sharply at hour 17 with a MAE of 23.16 gCO₂/kWh, followed by hour 16 at 19.94 gCO₂/kWh. The evening transition (16:00–18:00) is the most difficult forecasting period because solar generation falls to zero, residential demand rises, and the system often needs to dispatch additional conventional generation or increase imports. These are precisely the hours when CO₂ intensity transitions most rapidly between states, making the previous hour's value least representative of what follows.

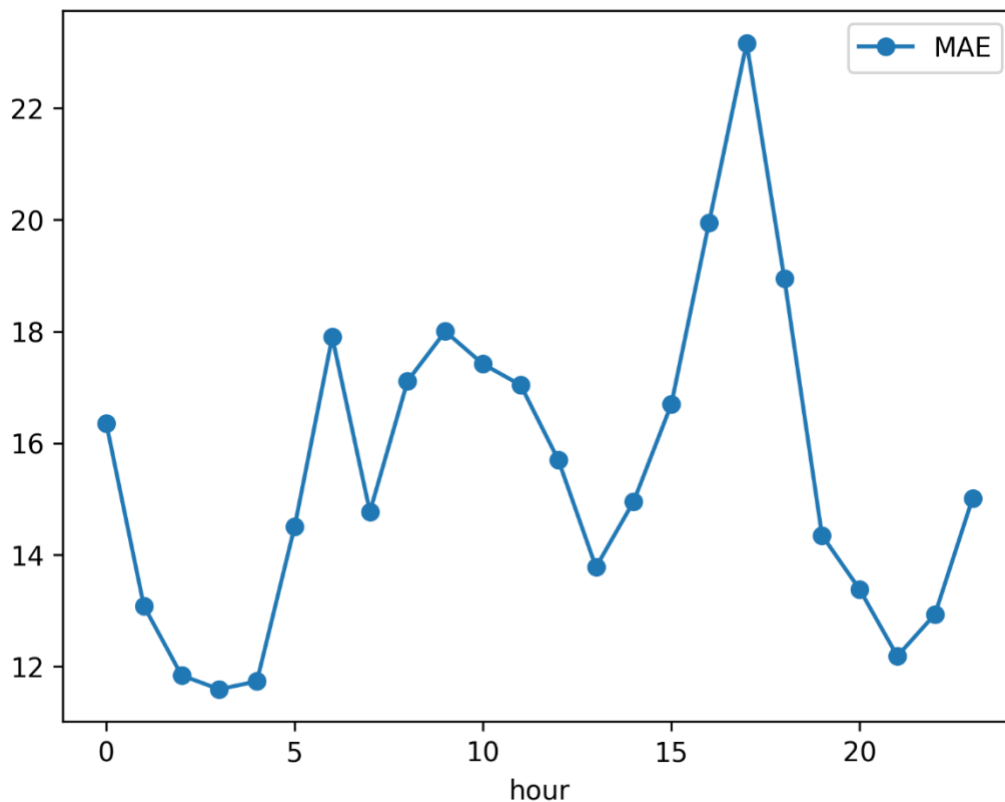


Figure 5.8. Mean absolute error (MAE) of the lag-1 persistence model for CO₂ emission intensity by hour of day across the 2024 test set. Errors are lowest during overnight hours (hour 3: 11.59 gCO₂/kWh) and peak sharply at hour 17 (23.16 gCO₂/kWh), corresponding to the evening transition when solar generation falls and conventional generation or imports increase.

By season: The seasonal pattern is the opposite of consumption. Winter is the easiest period to forecast (MAE: 11.14 gCO₂/kWh), while summer is the hardest (MAE: 19.25 gCO₂/kWh). In winter, strong and persistent wind keeps the system in stable low-carbon or high-carbon regimes for extended periods, making lag-1 a reliable predictor. Summer volatility is higher because solar generation introduces rapid intra-day swings in CO₂ intensity, midday solar pushes intensity down sharply, while early morning and evening periods see it rise again creating transitions that lag-1 persistence cannot anticipate. Spring sits at 17.06 gCO₂/kWh and autumn at 14.56 gCO₂/kWh.

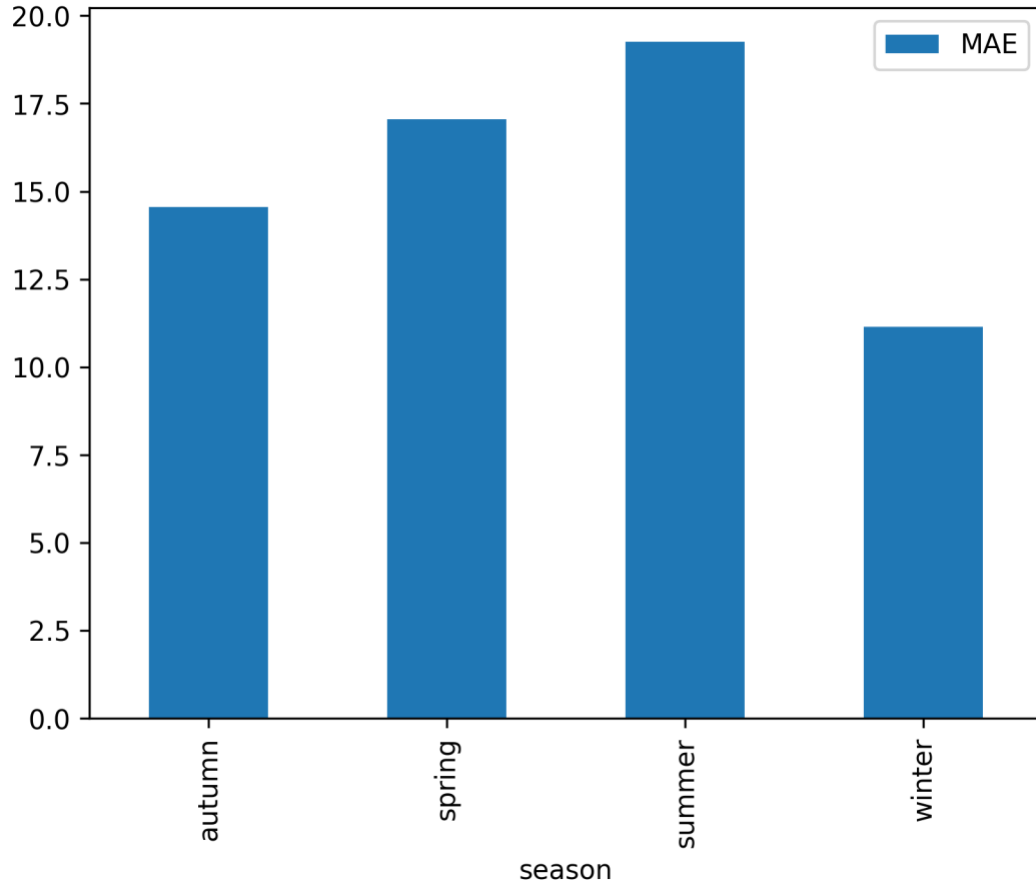


Figure 5.9. Mean absolute error (MAE) of the lag-1 persistence model for CO₂ emission intensity by season across the 2024 test set. Winter is the easiest season to forecast (11.14 gCO₂/kWh) while summer is the hardest (19.25 gCO₂/kWh), reflecting greater intra-day volatility from solar generation during summer months.

By renewable generation level: Conditional error analysis groups forecasts by terciles of lagged renewable generation (renewables_lag1). CO₂ forecasting errors decrease as renewable generation increases, the inverse of the consumption pattern. Low-renewable periods see a MAE of 19.22 gCO₂/kWh, medium-renewable 17.13 gCO₂/kWh, and high-renewable periods only 10.19 gCO₂/kWh. During high-renewable periods, CO₂ intensity is low and stable, once wind is blowing strongly, it tends to persist for several hours, and the lag-1 value captures this well. It is during transitional medium-renewable periods when the system could shift either direction - that CO₂ intensity is most unpredictable.

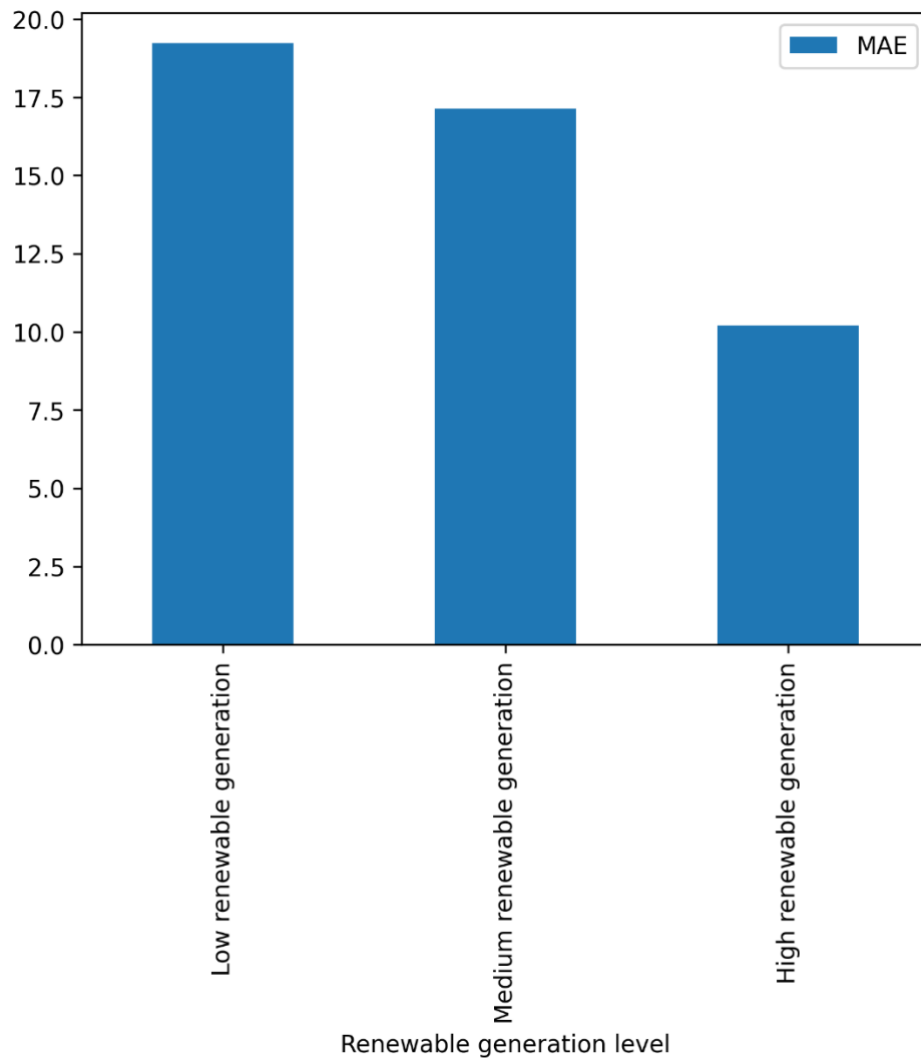


Figure 5.10. Mean absolute error (MAE) of the lag-1 persistence model for CO₂ emission intensity by lagged renewable generation level across the 2024 test set. Errors decrease as renewable generation increases - low (19.22 gCO₂/kWh), medium (17.13 gCO₂/kWh), and high (10.19 gCO₂/kWh) - reflecting greater intensity stability during sustained high-wind periods.

5.4 Summary of Results

Table 5.3 brings together the headline results across both targets.

Table 5.3. Summary of best model performance per target.

Target	Best model	Test MAE	MAE as % of mean	% vs lag-1
Consumption	Gradient boosting (tuned)	65.13 MWh	2.37%	+36.5%
CO ₂ intensity	Lag-1 persistence	15.52 gCO ₂ /kWh	15.06%	-

The two targets tell clearly different stories.

For electricity consumption, gradient boosting delivers a genuine and practically meaningful improvement at each step of increasing model complexity. Lag-1 persistence captures short-term autocorrelation. Linear regression adds structured effects of weather, calendar, and system state variables. Gradient boosting additionally captures the nonlinear interactions that a linear model cannot represent, most notably the U-shaped temperature relationship and the interplay between renewable output and cross-border flows. The 36.5% improvement over the lag-1 baseline and the reduction from 3.73% to 2.37% relative error are operationally significant at DK1's scale.

For CO₂ emission intensity, the lag-1 persistence model proves essentially unbeatable at the one-hour horizon. This finding is not a limitation of the modeling approach; it is a direct consequence of how CO₂ intensity behaves in a high-renewable system like DK1. Intensity is governed by instantaneous wind output, which changes rapidly and is not predictable from lagged system information alone. Once the previous hour's CO₂ value is available, it encodes the current renewable penetration state so efficiently that additional features provide no reliable improvement within a one-hour window.

In both cases, feature importance and SHAP results from the full models are persistence diagnostics, not answers to RQ3. The lag-1 target dominates so completely that all other predictors are effectively invisible. The no-lag-1 diagnostic setup is the correct basis for answering RQ3: for consumption, lagged production and exchange variables carry the most predictive weight when persistence is unavailable; for CO₂ intensity, it is lagged conventional generation and renewable generation that matter most, a result that aligns directly with the EDA findings in Chapter 3 and with the physical structure of DK1's generation mix. Taken together, these results provide direct empirical answers to all three research questions, with the sharpest and most actionable findings coming from the consumption target.

CHAPTER 6: DISCUSSION

6.1 Introduction

This chapter interprets the empirical results presented in Chapter 5 in the context of connecting the findings to the existing literature and reflects critically on the methodological choices and their implications. The chapter proceeds as follows. Section 6.2 addresses the three research questions directly. Section 6.3 examines the contrast between the two forecast targets, which is the most substantive finding of the study. Section 6.4 discusses the implications of strong persistence for feature importance interpretation. Section 6.5 connects the findings to prior work. Section 6.6 discusses practical implications for grid operators. Section 6.7 acknowledges the study's limitations and suggests directions for future work.

6.2 Interpreting the Results

The results give a clear but asymmetric answer to the study's core question.

For electricity consumption, the answer is yes on both counts. A tuned gradient boosting model reduces the typical hourly forecast error from 102.63 MWh to 65.13 MWh - a 36.5% improvement over the lag-1 persistence baseline - bringing the relative error down to 2.37% of average consumption. This is a practically useful level of accuracy for a one-hour-ahead task, and the improvement is not a statistical artifact: it comes in measurable steps. Linear regression alone, by incorporating weather, calendar, and lagged system variables into a linear framework, already cuts error by 16.9% over persistence. Gradient boosting then adds a further 23.7% improvement on top of that by capturing the nonlinear and interactive structure that a linear model cannot represent. Each layer of complexity earns its place.

For CO₂ emission intensity, the picture is reversed. Neither linear regression nor gradient boosting improves over the lag-1 persistence baseline in MAE terms. The tuned gradient boosting model achieves a MAE of 15.70 gCO₂/kWh - marginally worse than the 15.52 of the lag-1 baseline. The added complexity of gradient boosting is not justified for this target at this horizon. The simple persistence model is the best available one-hour-ahead forecast for CO₂ intensity given the information set used here.

As for which variables drive the forecasts - the answer depends entirely on which model specification is examined. In the full predictive setup, the lag-1 target dominates both models so completely that all other predictors are effectively irrelevant. The sensitivity analysis gives a more interpretable picture, and the results - discussed in detail in Section 5.4 - are physically coherent with DK1's generation mix.

6.3 The Contrast Between Targets

The most striking result in this thesis is the asymmetry between the two forecast targets, and it deserves careful interpretation rather than being passed over as a negative result.

Electricity consumption is a quantity shaped by human behavior - heating, working hours, cooking, lighting - embedded in a stable weekly and daily rhythm. This rhythm means that the previous hour's consumption is not the whole story: knowing what time of day it is, what season it is, and what the weather is adds genuine predictive value beyond what the lag-1 value already contains. Gradient boosting can exploit these additional signals, particularly the nonlinear relationship between temperature and consumption and the interactions between time of day and renewable output. The result is a model that meaningfully outperforms persistence and linear regression.

CO₂ intensity is a fundamentally different quantity. It is not shaped by human behavioral rhythms in the same way - it is determined primarily by the instantaneous renewable generation state, which is governed by wind and solar availability. Wind availability in DK1 changes on timescales driven by atmospheric dynamics, not by the clock. The ACF analysis in Chapter 3 reflects this: CO₂ intensity has a lag-1 autocorrelation of 0.947 but drops sharply to 0.510 at lag-24 and only 0.227 at lag-168, compared to consumption's 0.822 and 0.815 at those same lags. CO₂ intensity stays in the same regime - high wind, low intensity, or low wind, high intensity - for hours at a time, but those regimes do not follow a predictable daily or weekly cycle. Once you know last hour's CO₂ intensity, you already know the current renewable penetration state as accurately as any lagged weather or system variable can tell you. This is why the lag-1 value is so hard to beat.

This contrast has a direct methodological implication. The task of forecasting CO₂ intensity one hour ahead is fundamentally a task of predicting short-term changes in renewable generation. At a one-hour horizon, and without access to contemporaneous wind output or a numerical weather forecast for the next hour, the information required to substantially improve on persistence is not fully available in the feature set. This is not a weakness of gradient boosting - it is a ceiling imposed by what can be known at forecast time.

6.4 Persistence Effects and Feature Importance

A recurring theme across both targets is the dominance of the lag-1 value. This dominance has important implications for how the feature importance and SHAP results should be read.

When a model includes a highly autocorrelated lag-1 predictor, the gradient boosting algorithm rationally allocates most of its predictive power to that variable. It would be inefficient to do otherwise - the lag-1 value already explains the vast majority of the variance. The consequence is that gain-based importance scores and SHAP values from the full predictive model are almost entirely a reflection of

autocorrelation structure, not of the substantive drivers of the system. Interpreting the high importance of `cons_lag_1` or `co2_lag_1` as evidence that "past values cause future values" would be circular and treating the low importance of weather or exchange variables as evidence that they do not matter for the system would be misleading.

The sensitivity analysis - the no-lag-1 setup - is specifically designed to address this. By removing the dominant predictor, the remaining features are forced to compete on equal footing, and the importance rankings that emerge are more informative about which observable variables carry systematic predictive information about the next hour's state. For consumption, the emergence of lagged net exchange and total production as leading predictors makes physical sense: in DK1, the balance between domestic generation and cross-border flows is the most direct observable signal of current system demand, and its one-hour lag is the best available proxy for what the system looks like right now. For CO₂ intensity, the emergence of `conventional_lag1` and `renewables_lag1` as the dominant predictors is physically coherent with the DK1 generation mix - when wind exceeds approximately 2,000 MWh, CO₂ intensity approaches to near zero; when it is low, intensity rises sharply. The sensitivity analysis is recovering that same relationship from the lagged data.

It is important to note that the sensitivity analysis results should not be interpreted causally either. High SHAP values for `conventional_lag1` and `renewables_lag1` do not mean that conventional or renewable generation causes CO₂ intensity - rather, they mean these lagged values are informative correlates of next-hour intensity in a way that the model can exploit. The distinction between prediction and causation is maintained throughout.

6.5 Connections to the Literature

The consumption forecasting results are broadly consistent with the literature reviewed in Chapter 2. Consistent with (Chen et al., 2023), tree-based ensemble models can achieve lower forecast errors than linear regression. The U-shaped temperature-consumption relationship identified in the EDA suggests a source of nonlinearity that linear models cannot adequately capture, though its contribution is obscured in the full model SHAP by the dominance of `cons_lag_1`. The magnitude of improvement over the lag-1 baseline - 36.5% in MAE - likely reflects DK1's pronounced nonlinear weather-demand dynamics, and it is plausible that this drives a larger portion of gradient boosting's advantage here than in systems with more moderate weather exposure.

The CO₂ intensity results are harder to benchmark directly, because the existing literature on machine learning for CO₂ forecasting is sparse and operates at different horizons and in different system contexts. (Leerbeck et al., 2020) achieved NRMSEs ranging from 0.095 to 0.183 for average CO₂ intensity forecasting in DK2 using a linear hybrid model - a result that cannot be directly compared to the present study given differences in target definition, horizon, and system. (Maji et al., 2023) showed that deep learning models can produce accurate multi-day carbon intensity forecasts across selected regions in the US, Europe, and Australia, with 4-day MAPEs ranging from 4.80% to 13.93% across six regions. Multi-day forecasting is however a fundamentally different task from one-hour-ahead, where the lag-1 value is far more dominant. The finding that lag-1 persistence is essentially unbeatable at the one-hour horizon is not inconsistent with these results - they simply operate at longer horizons where persistence degrades more rapidly. Whether the conclusion holds at two-hour or four-hour horizons is an open question.

The literature also supports the methodological concern that feature-importance results must be interpreted cautiously when targets and predictors are strongly autocorrelated or correlated with each other. (Lundberg & Lee, 2017) explicitly note that SHAP values describe model behavior rather than causal structure, and (Strobl et al., 2007) demonstrate that tree-based variable-importance measures can be unreliable when predictors differ in scale of measurement, number of categories, or number of possible split points. Both concerns are relevant to the full predictive models in this study, where the

lag-1 target variable is highly autocorrelated and strongly related to other lagged predictors. The sensitivity analysis approach adopted here is one practical response to this challenge, although it still has its own interpretive limitations.

6.6 Practical Implications

The consumption forecasting results have direct relevance for grid operators such as Energinet. A model that reduces typical hourly forecast error from 102.63 MWh to 65.13 MWh - a reduction of approximately 37.5 MWh per hour - may be relevant for short-term operational planning, although direct effects on reserve requirements are not quantified in this thesis. At DK1's scale, where peak consumption can exceed 3,500 MWh and interconnector flows are actively managed across five borders, a reduction in forecast uncertainty of this magnitude has operational value. The conditional performance analysis adds nuance: the model performs best during overnight hours and summer months, and worst during the morning and evening peak transitions and during high-renewable periods. Grid operators should expect larger forecast errors precisely when the system is under the most dynamic conditions - the morning ramp-up, the evening peak, and periods of high wind variability - which are also the periods where accurate forecasts are most valuable. This is a limitation worth acknowledging explicitly when deploying the model operationally.

For CO₂ intensity, the practical implication is more nuanced. The fact that lag-1 persistence is the best available one-hour-ahead forecast does not mean CO₂ intensity is unforecastable - it means it is forecastable at the level of short-term persistence. For real-time carbon accounting, demand-response signaling, and carbon pricing mechanisms, a lag-1 persistence forecast may still be operationally useful, particularly during stable high-renewable periods where CO₂ intensity is low and relatively constant. The conditional analysis showed that forecasting errors for CO₂ intensity are lowest precisely during high-renewable periods (MAE: 10.19 gCO₂/kWh) and during winter (MAE: 11.14 gCO₂/kWh) - the conditions under which carbon-aware demand response is most likely to be triggered. This is a practically meaningful finding: while the models cannot reliably predict when the next intensity spike will occur, they can provide reasonably accurate forecasts precisely during the clean-grid conditions that matter most for carbon-aware consumption strategies.

6.7 Limitations and Future Work

Several limitations of this study should be acknowledged.

The most important is the treatment of weather variables as known at forecast time. Throughout this study, observed historical weather data is used during both training and evaluation. In an operational setting, weather inputs would need to come from a numerical weather forecast, which would introduce additional forecast error. The performance figures reported here therefore represent what is achievable under favorable conditions - when short-term weather forecasts are accurate - rather than a fully operational deployment scenario. Future work should evaluate model performance under realistic weather forecast uncertainty, for example by using actual weather forecast output as input rather than observed values.

A related limitation concerns the feature set for CO₂ intensity forecasting. The inability to beat lag-1 persistence at a one-hour horizon is partly a consequence of the features available: without direct access to the current hour's wind output or a numerical forecast of next-hour wind speed, the model cannot anticipate the rapid transitions in CO₂ intensity that occur when wind generation changes sharply. Including a weather forecast as a feature - rather than observed weather - and extending to multi-hour horizons where persistence degrades more rapidly would both be promising extensions.

The study covers the period January 2022 to December 2024. This is a recent and relevant period, but it encompasses only three years, which means that some rare but important system conditions - extreme weather events, major generation outages, market disruptions - may be underrepresented in the training data. Model generalization to conditions outside this window is not guaranteed. The study also uses a single chronological test year. Future work could use rolling-origin evaluation across multiple periods to assess robustness across different weather and market conditions.

The modeling approach is also limited in scope by design. Only gradient boosting, linear regression, and naïve persistence are compared. Deep learning methods such as LSTM or Temporal Fusion Transformers have shown strong performance in multi-step energy forecasting tasks and could provide a useful comparison, particularly for CO₂ intensity where the gradient boosting model fails to beat the persistence baseline. Probabilistic forecasting - which produces a distribution over future values rather than a point estimate - would also be a natural extension, given the operational value of uncertainty quantification for reserve scheduling.

Finally, the analysis is restricted to the DK1 bidding zone with weather observations from a single station in Western Jutland. While this ensures spatial consistency, it means that the models do not capture spatial variation in weather across the zone, and the findings may not generalize directly to other bidding zones or other high-renewable systems without retraining.

Despite these limitations, the findings are supported within the empirical setup used in this thesis, but their robustness across other periods, zones, and system conditions remains a topic for future work. The benefit of gradient boosting for consumption forecasting is supported by a large test set (8,784 hourly observations), a systematic three-way model comparison, and a clear mechanistic explanation rooted in the nonlinear structure of the DK1 system. The failure to improve on lag-1 persistence for CO₂ intensity is equally robust and is a finding that, while perhaps unexpected, is well-grounded in the autocorrelation structure of the series and the nature of what can be known at a one-hour forecast horizon.

6.8 Summary

This chapter has interpreted the empirical results in the context of the three research questions and the broader literature. The central finding is that gradient boosting improves substantially on persistence for consumption but not for CO₂ intensity, and its implications are discussed in detail above and summarized in Chapter 7. This contrast is not a methodological failure - it reflects a fundamental difference in the nature of the two quantities and in what information is realistically available at forecast time. Feature importance results from the full predictive models should be read with caution, as they primarily reflect autocorrelation structure rather than substantive system drivers. The sensitivity analysis provides a more interpretable picture of which exogenous variables matter, and its results are physically coherent with the structure of DK1's generation mix. The practical relevance of the consumption forecasting results is clear; the implications for CO₂ intensity forecasting are more nuanced but still meaningful for carbon-aware demand management during stable high-renewable periods.

CHAPTER 7: CONCLUSION

This thesis set out to develop and evaluate machine learning models for one-hour-ahead forecasting of electricity consumption and CO₂ emission intensity in the DK1 bidding zone of Western Denmark, and to assess whether gradient boosting offers meaningful improvements over simpler baselines. Three years of hourly data (2022–2024) from Energinet and DMI were used to train and evaluate naïve persistence, linear regression, and gradient boosting models on a held-out 2024 test set.

The central finding is an asymmetry between the two targets. For electricity consumption, gradient boosting performs meaningfully better than the lag-1 persistence baseline - reducing MAE by 36.5% and bringing the relative error down to 2.37% of average consumption. This improvement is not simply a product of adding more variables; it suggests that gradient boosting can exploit nonlinear relationships and interactions more effectively than the linear model. Each step up in model complexity - from persistence to linear regression to gradient boosting - adds predictive value.

For CO₂ emission intensity, the result is different and negative in predictive terms. No trained model improves on the lag-1 persistence baseline in MAE terms at the one-hour horizon. The gradient boosting model achieves a MAE of 15.70 gCO₂/kWh against the baseline's 15.52 - a marginal deterioration. This result indicates that lag-1 persistence is an extremely strong benchmark for one-hour-ahead CO₂ intensity forecasting under the leakage-controlled feature setup used in this thesis. Rather than being treated as a failed result, this finding is scientifically useful because it shows that short-term CO₂ intensity in DK1 is difficult to improve beyond persistence with the tested information set.

The feature importance analysis reinforces this picture. In both full predictive models, the lag-1 target dominates to such a degree that the results say more about autocorrelation structure than about the system's physical drivers. The sensitivity analysis - which excludes the lag-1 target - is more informative on this front: for consumption, lagged production and exchange variables emerge as the most important exogenous predictors; for CO₂ intensity, it is lagged conventional generation and renewable generation. Both findings are physically coherent with DK1's generation mix, and the dynamics described in the exploratory analysis.

The results answer the three research questions directly. RQ1: gradient boosting achieves a test MAE of 65.13 MWh (2.37% of average consumption) for electricity consumption, and lag-1 persistence at 15.52 gCO₂/kWh is the best achievable forecast for CO₂ intensity at this horizon. RQ2: for consumption, each step up in model complexity adds genuine value - linear regression improves 16.9% over persistence, gradient boosting a further 23.7% - while for CO₂ intensity no trained model improves over persistence in MAE terms. RQ3: full model feature importance reflects autocorrelation structure rather than system drivers; the diagnostic no-lag-1 setup shows that lagged production and exchange variables drive consumption forecasts while lagged conventional generation and renewable generation drive CO₂ intensity forecasts when persistence is unavailable. The contrast between the two targets - strong improvement for consumption, persistence unbeatable for CO₂ - is itself a substantive finding that reveals what is and is not forecastable in a high-renewable system at a one-hour horizon.

Several directions for future work follow naturally from these findings. The most immediate is the evaluation of model performance under realistic weather forecast conditions, rather than observed weather - the current setup approximates an upper bound on accuracy that a deployed system would not achieve. Extending the forecasting horizon beyond one hour would also be informative: at longer horizons, the lag-1 value degrades faster, and gradient boosting may provide more substantial improvements for CO₂ intensity. Including probabilistic forecasting - providing confidence intervals rather than point estimates - would be relevant for reserve planning and carbon-aware decision-making, although operational value would require further validation. Finally, exploring whether the CO₂ intensity findings generalize to other high-renewable bidding zones, or whether DK1's specific combination of wind dominance and cross-border interconnection makes it a distinctive case, remains an open question worth investigating.

Overall, this thesis contributes a grounded empirical evaluation of machine learning forecasting in one of Europe's most renewable-intensive electricity systems. The contrast between the two targets is itself the central empirical contribution: it shows precisely where model complexity is justified and where it is not and reveals something specific about the predictability of consumption versus CO₂ intensity in a wind-dominated system. The two-setup framework - separating predictive performance evaluation from driver analysis - offers a practical template for similar forecasting problems in other high-renewable contexts.

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Author Contribution

This thesis grew out of a joint effort from the very beginning. We developed the research questions and overall direction together through ongoing discussion. Wahid took the lead on building the data foundation: sourcing, cleaning, and integrating the three datasets, and conducting the initial exploratory analysis including feature engineering and autocorrelation work. Sifat built on this foundation to develop the modeling framework, implementing the gradient boosting pipeline, running the hyperparameter search, and carrying out the SHAP and conditional performance analysis. From there, the work became increasingly collaborative - we interpreted the results together, wrote chapters in parallel, and revised each other's drafts throughout. The final structure, arguments, and conclusions reflect decisions we made jointly. Both authors read and approved the final manuscript.